

A cognitive perspective on trust in science (PhD thesis)

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Preface

This is a Quarto book.

To learn more about Quarto books visit <https://quarto.org/docs/books>.

1 Introduction

This is a book created from markdown and executable code.

See (**knuth84?**) for additional discussion of literate programming.

2 Summary

In summary, this book has no content whatsoever.

3 Quasi-universal acceptance of basic science in the US

Many people report a low degree of trust in science, or endorse conspiracy theories that violate basic scientific knowledge. This might indicate a wholesale rejection of science in substantial segments of the population. In four studies, we asked 782 US participants questions about trust in science, conspiracy beliefs, and basic science (e.g. the relative size of electrons and atoms). Participants were provided with the scientifically consensual answer to the basic science questions, and asked whether they accept it. Acceptance of the scientific consensus was very high in the sample as a whole (95.1%), but also in every sub-sample (e.g., no trust in science: 87.3%; complete endorsement of flat Earth theory: 87.2%). This quasi-universal acceptance of basic science suggests that people are motivated to reject specific scientific beliefs, and not science as a whole. This could be leveraged in science communication.

i available as a preprint here:

Pfänder, J., Kerzreho, L., & Mercier, H. (2025). *Quasi-universal acceptance of basic science in the US*. https://doi.org/10.31219/osf.io/qc43v_v2

For supplementary materials, please refer to the preprint.

3.1 Introduction

Trust in science is related to many desirable outcomes, from acceptance of anthropogenic climate change (Cologna and Siegrist 2020) or vaccination (Sturgis, Brunton-Smith, and Jackson 2021; Lindholt et al. 2021) to following recommendations during COVID (Algan et al. 2021, which suggests that trust in science was the most important predictor of these behaviors).

Although recent global evidence shows that trust in science is moderately high (Cologna et al. 2024), it is far from being at ceiling. Large-scale polls have shown that people who report a high degree of trust in science are a minority in most countries, and they are outnumbered by people who have low trust in science in many areas, e.g., most of Africa and significant parts of Asia (Wellcome Global Monitor 2018, 2020). Moreover, trust in science has recently been

declining in some countries (Algan et al. 2021; Brian and Tyson 2023; although see Wellcome Global Monitor 2021; and Funk and Kennedy 2020) and in the US it is increasingly polarizing (Gauchat 2012; Krause et al. 2019; Li and Qian 2022).

Besides low answers on general trust in science questions, another indicator of distrust in science is the belief in conspiracy theories that question the scientific consensus on issues such as vaccination, climate change, and even the shape of the Earth. Conspiracy theories—in the realm of science or elsewhere—typically accuse a small group of powerful people to pursue nefarious goals in secrecy (Douglas et al. 2019; Mede and Schäfer 2020). Some of these conspiracy theories are widespread (Rutjens and Većkalov 2022). In 2023, a survey in eight different countries found that up to 24% of respondents agreed that “climate change is a hoax and scientists touting its existence are lying” (Stockemer and Bordeleau 2024). Similarly, in 2021, 40% of Americans believed that “the dangers of genetically-modified foods are being hidden from the public” (down from 45% in 2020, Uscinski et al. 2022). People who hold such views not only reject the relevant scientifically consensual facts, but also tend to believe in other conspiracy theories (Lewandowsky, Gignac, and Oberauer 2013; Hornsey, Harris, and Fielding 2018a, 2018b), and tend to say that they distrust science more generally (Vranic, Hromatko, and Tonković 2022; Stockemer and Bordeleau 2024). In spite of these correlations, the causal relationship between declaring general distrust in science and believing in one or several anti-science conspiracy theories is not clear. Although conspiracy thinking has been identified as a “root cause” of anti-science attitudes (Hornsey and Fielding 2017) in the past, these claims rest largely on observational data.

What does this apparent lack of trust in science actually entail? Do people who say they do not trust science, or who believe in conspiracy theories at odds with well-established science, reject most of science? Or, on the contrary, do they object to a few specific facets of science, while still accepting the overwhelming majority of basic science?

A common conception of trust is a willingness to be vulnerable to another party, whether an individual, a group, or an institution (Mayer, Davis, and Schoorman 1995; Rousseau et al. 1998). Accordingly, trust in science has been defined as “one’s willingness to rely on science and scientists (as representatives of the system) despite having a bounded understanding of science” (Wintterlin et al. 2022, 2). Past research has disentangled this general concept of trust in science in various ways. Some research has identified different components of trust, the number of which varies, but which generally cover an epistemological and ethical dimension (Wilholt 2013; Intemann 2023). For example, Hendriks, Kienhues, and Bromme (2015) suggest distinguishing between expertise/competence, integrity, and benevolence, while Besley, Lee, and Pressgrove (2021) add openness. Other research has highlighted differences in trust between scientific disciplines (Altenmüller, Wingen, and Schulte 2024; Gligorić, Kleef, and Rutjens 2024; Gauchat and Andrews 2018). However, little research has assessed trust in specific scientific findings, besides contentious topics such as vaccines (e.g. Hornsey, Harris, and Fielding 2018a), climate change (e.g. Stockemer and Bordeleau 2024), evolution (e.g. Nadelson and Hardy 2015), genetically modified organisms (GMOs) (e.g. Fernbach et al. 2019), or a combination of such topics (see e.g. Lewandowsky, Gignac, and Oberauer 2013). To the best

of our knowledge, no research has investigated the extent to which people trust basic science facts (e.g. electrons are smaller than atoms). An extensive literature on science literacy has assessed whether people know such facts (National Academies of Sciences, Engineering, and Medicine 2016), but not whether they accept the facts once presented to them.

Why does it matter whether trust in science is or is not related to trust in specific scientific findings? First, this question has theoretical implications. According to the most prominent explanation of (dis)trust in science—the deficit model—science knowledge is the main driver of attitudes towards science in general. A prediction of this model is that average trust in science on specific facts should be strongly associated with general trust in science. By contrast, a disconnect between the two would be in line with motivated reasoning accounts of trust in science. According to these accounts, science rejection serves to maintain coherence with other beliefs or behaviors (Lewandowsky and Oberauer 2016; Hornsey 2020). For example, someone might say they do not trust science in general because they are skeptical towards vaccines, not because they actually distrust most of science. A prediction of the motivated reasoning account is that general trust in science should be strongly correlated with conspiracy beliefs. Second, the present question has practical implications: Many communication attempts leverage the scientific consensus (e.g. on vaccination, climate change, etc., for review, see Van Stekelenburg et al. 2022; see also Većkalov et al. 2024). These attempts are more likely to be successful if everyone trusts basic science than if some people reject science wholesale.

3.1.1 The present studies

In a series of four pre-registered online studies (total $n = 782$), we asked US participants questions about well-established, consensual scientific facts. For each question, we asked participants what they thought the correct answer was (testing their knowledge of science), we informed them of the scientifically accepted answer, and asked them whether they accepted it (measuring their trust in basic science). We also measured participants’ trust in science using standard measures, as well as their beliefs in various conspiracy theories and their tendency to engage in conspiratorial thinking. The four studies, including materials, hypotheses, and analyses, were pre-registered and all materials and data are accessible via the Open Science Framework (<https://osf.io/8utsj/>). The differences between the four studies are summarized presently, and the methods are detailed below.

3.1.1.1 Materials

In Study 1 (data collected on March 5, 2024), we used questions drawn from questionnaires of scientific knowledge (e.g. “Are electrons smaller, larger, or the same size as atoms? [Smaller; Same size; Larger]”), supplemented by a ‘trick’ question (“Where do trees mainly draw the materials with which they create their mass? [Earth; Water; Air]”; correct answer: Air). In Studies 2 (data collected on April 3, 2024) and 3 (data collected on April 22, 2024), this last question was removed. The scientific facts used in Studies 1 to 3 represent long-established

and basic knowledge. In Study 4 (data collected on August 13, 2024) we used more recent, much less basic scientific discoveries (e.g. “What is the electric charge of the Higgs Boson, as established in 2012? [$1.602176634 \times 10^{-19}$; 0; $3.2 \times 10^{-19}\text{C}$]”; correct answer: 0, i.e. electrically neutral).

3.1.1.2 Presentation of the scientific consensus

In Study 1, we simply told participants that they would be provided with the scientifically consensual answer. However, for participants to accept this answer, they must not only trust science, but also trust that we are presenting them with the actual scientifically consensual answer. To remove this issue, in Studies 2 to 3, we presented participants with a short explanation of the correct answer, as well as links to three sources per answer (e.g. Wikipedia, National Geographic or NASA). In Study 4, as the topics were more complex, we did not provide an explanation, but still provided two sources per answer.

3.1.1.3 Measure of acceptance of the scientific consensus

In Study 1, we simply looked at whether participants accept the scientifically consensual answer or not. In the subsequent studies, we asked participants to explain cases in which they disagreed with the scientific consensus. This revealed that a number of participants had made a mistake (misunderstanding, selecting the wrong answer). As a result, in Studies 3 and 4, participants who have indicated that they rejected the scientifically consensual answer were offered the option to revise their answer, or to keep rejecting it.

3.1.1.4 Additional questions

In Studies 3 and 4, we attempted to understand how some people who say they do not trust science still accept scientifically consensual answers, by asking them whether they accepted the answers on the basis of trust in science or because they had independently verified them.

3.1.1.5 Samples

Studies 1 and 2 were conducted on the standard sample of US participants recruited on the platform Prolific Academic. In order to increase the share of participants with low trust in science, and who endorse conspiracy theories, Studies 3 and 4 used the same platform, but only recruited participants who had declared previously being skeptical of vaccination.

3.1.1.6 Hypotheses

The main goal of the present studies is descriptive: to find out whether participants who report not trusting science, or who believe in conspiracy theories, still accept most well-established scientific facts. However, based on our literature review, we also tested two directional hypotheses (pre-registered as research questions in the Study 1):

H1: Higher trust in science is associated with more science knowledge and more acceptance of the scientific consensus

H2: Higher conspiracy thinking/belief is associated with less science knowledge and less acceptance of the scientific consensus

3.2 Methods

3.2.1 Deviations from preregistration

For Study 2, we restricted our main hypotheses about acceptance to cases in which participants initially provided a wrong answer. However, this meant the more participants had initially provided correct answers, the fewer opportunities they had for accepting correct answers. We provide results on these conditional correlations—for Study 2 and for all other studies—in the ESM ¹. However, for the analysis presented here, we proceeded as preregistered for all other studies, by reporting unconditional correlations between acceptance and trust in science, or, respectively, conspiracy belief.

3.2.2 Procedure

After providing their consent to participate in the study, participants were given an attention check “While watching the television, have you ever had a fatal heart attack?” [1-6; 1 = Never, 6 = Often]. All participants who did not answer “1 = Never” were excluded. Participants then read the following instructions: “We will ask you 10 questions about science. After each question, we will provide you with the scientifically consensual answer and ask whether you accept it.” Next, participants answered a set of 10 basic science questions in random order. After each question, participants were presented with an answer reflecting the scientific consensus, and asked whether they accepted it. In Studies 2 and 3, participants additionally saw a short explanation, partly based on explanations generated by ChatGPT, and three links to authoritative sources supporting the answer. In Study 4, we provided only two links and no

¹The two median values for all information and for the subset of impressive information are not not exactly the same, but very similar, because participants rated most knowledge elements as impressive, making the general recall score and the recall score for impressive elements very similar.

explanation. Participants then answered questions on conspiracy thinking, conspiracy beliefs, and trust in science.

In Studies 2, 3, and 4, we presented participants with open-ended questions so they could explain their rejection of the scientific consensus. In Studies 3 and 4, we additionally gave participants the option to change their answer and accept the scientific consensus. Finally, at the end of Studies 3 and 4, we asked participants: “For the questions in which you agreed with the scientific consensus, would you say that...?” The answer options were: (i) “You mostly agree with the consensus because, on that question, you trust scientists”, (ii) “You mostly agree with the consensus because you have been able to independently verify it”, and (iii) “Other”, with a text box for participants to explain. Participants who selected “You mostly agree with the consensus because you have been able to independently verify it”, were asked the open-ended follow-up question: “Could you please tell us how you independently verified the information?”.

3.2.3 Participants

After removing failed attention checks, the total sample size was 782 (194 in Study 1, six failed attention checks; 190 in Study 2, 11 failed attention checks; 200 in Study 3, no failed attention checks; 198 in Study 4, two failed attention checks) participants in the US, recruited through Prolific. Details and demographics can be found in the online supplemental material. While samples for Studies 1 and 2 were convenience samples, Studies 3 and 4 were conducted on a sample holding vaccine-skeptic beliefs. Prolific allows selecting participants based on their answers to a range of questions. We picked three of these questions and only recruited participants who met our criteria for each of them:

1. “Please describe your attitudes towards the COVID-19 (Coronavirus) vaccines: [For (I feel positively about the vaccines); Against (I feel negatively about the vaccines); Neutral (I don’t have strong opinions either way); Prefer not to say]”. We selected participants who answered “Against”.
2. “Have you received a coronavirus (COVID-19) vaccination? [Yes (at least one dose); No; Prefer not to answer]”. We select only people who answered “No”.
3. “On a scale from 1-7, please rate to what extent you agree with the following statement: I believe that scheduled immunizations are safe for children. [1 (totally disagree); 2 (disagree); 3 (somewhat disagree); 4 (neither agree nor disagree); 5 (somewhat agree); 6 (agree); 7 (totally agree); rather not say]”. We select only people who answered “1”, “2”, or “3”.

3.2.4 Materials

3.2.4.1 Scientific facts

Studies 1 to 3 used 10 facts drawn from widely used questionnaires about science knowledge (Allum et al. 2008; Durant, Evans, and Thomas 1989; Miller 1998) sometimes referred to as the “Oxford scale” (Gauchat 2011). A ‘trick’ question was added in Study 1 and removed as its wording proved unclear. Study 4 used 10 more recent scientific discoveries. Table 3.1 shows all questions and their answer options.

Table 3.1: Science knowledge items

Study 1-3	Study 4
1 Do antibiotics kill viruses as well as bacteria? [Yes, both; No, only viruses; No, only bacteria]	For which disease is the drug bedaquiline, developed in 2007, a treatment? [Tetanus; Tuberculosis; Malaria]
2 Are electrons smaller, larger, or the same size as atoms? [Smaller; Same size; Larger]	What is the maximum speed a proton can attain in the largest particle collider as to 2015? [90% of the speed of light; 99% of the speed of light; the speed of light]
3 Have the continents on Earth been moving for millions of years or have they always been where they are now? [They have been moving; They have always been where they are now]	Kepler-452b is an exoplanet revolving around the star Kepler-452. How far away from the star is it, as established by astronomers in 2015? [97 million mi; 1,2 million mi; 1254 million mi]
4 What decides whether a baby is a boy or a girl ? Is it the father’s genes, the mother’s genes, or both? [The mother’s genes; the father’s genes; both]	Using bomb-pulse dating with carbon 14, what is the age of the oldest known vertebrate, as established in 2016? [138 years; 205 years; 392 years]
5 Do lasers work by focusing sound waves? [Yes; No]	How many more glial cells are there in the brain in comparison with neurons, as established in 2016? [The same amount; Twice as many; Ten times as many]
6 How long does it take for Earth to go around the sun: one day, one month, or one year? [One day; One month; One year]	As predicted by the general theory of relativity, how many times would the Earth keep orbiting if the Sun disappeared, as established in 2012? [47 seconds; 8 minutes; 2 hours]
7 Are diamonds made of carbon? [Yes; No]	What is the electric charge of the Higgs Boson, as established in 2012? [$1.602176634 \times 10^{-19}$; 0; $3.2 \times 10^{-19}C$]
8 Which travels faster : light or sound? [Light; Sound]	What is the age of the oldest materials formed on Earth, as established in 2020? [Less than 4.6 Ga; Around 4.6 Ga; More than 4.6 Ga]
9 Is common table salt made of calcium carbonate? [Yes; No]	With the best current cloning techniques, what is the average success rate when operated on mice, as of 2010? [2,7%; 9,4%; 17,2%]
10 Is water made of molecules containing one oxygen and two hydrogen atoms? [Yes; No]	What was the strength of the Earth magnetic field 3.7 billion years ago, as discovered this year? [15 microtesla; 30 microtesla; 45 microtesla]

- 11 *Where do trees mainly draw the materials
with which they create their mass? [Earth;
Water; Air]

* Only used in Study 1

3.2.4.2 Conspiracy beliefs

We selected 10 science/health related conspiracy theories from the Belief in Conspiracy Theory Inventory (BCTI) (Pennycook, Binnendyk, and Rand 2022) (Table 3.2). Participants were asked: “Below is a list of events for which the official version has been disputed. For each event, we would like you to indicate to what extent you believe the cover-up version of events is true or false. [1-9; labels: 1 - completely false, 5 - unsure, 9 - completely true]”.

3.2.4.3 Conspiracy thinking

For all results presented here, we used the four-item conspiracy mentality questionnaire (CMQ) (Bruder et al. 2013). We also assessed the single item conspiracy beliefs scale (SICBS) (Lantian et al. 2016). Details and comparisons between the scales can be found in the ESM.

Table 3.2: Conspiracy items.

1	The Apollo moon landings never happened and were staged in a Hollywood film studio.
2	A cure for cancer was discovered years ago, but this has been suppressed by the pharmaceutical industry and the U.S. Food and Drug Administration (FDA).
3	The spread of certain viruses and/or diseases is the result of the deliberate, concealed efforts of vested interests.
4	The claim that the climate is changing due to emissions from fossil fuels is a hoax perpetrated by corrupt scientists who want to spend more taxpayer money on climate research.
5	The Earth is flat (not spherical) and this fact has been covered up by scientists and vested interests.
6	There is a causal link between vaccination and autism that has been covered up by the pharmaceutical industry.
7	In the 1950s and 1960s more than 100 million Americans received a polio vaccine contaminated with a potentially cancer-causing virus.
8	Proof of alien contact is being concealed from the public.
9	Hydroxychloroquine has been demonstrated to be a safe and effective treatment of COVID and this information is being suppressed.
10	Dinosaurs never existed, evolution is not real, and scientists have been faking the fossil record.

3.2.4.4 Trust in science

In all analyses reported in the main paper, we measure trust in science via a question selected from the Wellcome Global Monitor surveys (Wellcome Global Monitor 2018, 2020): “In general,

would you say that you trust science a lot, some, not much, or not at all? [1 = Not at all, 2 = Not much, 3 = Some, 4 = A lot]”. We chose this question as it seemed to be the most general one. In the ESM, we additionally report results for two alternative measures of trust included in our studies: Another from the WGM surveys (“How much do you trust scientists in this country? Do you trust them a lot, some, not much, or not at all? [1 = Not at all, 2 = Not much, 3 = Some, 4 = A lot]”), and one from the Pew Research Center (e.g. Funk, Johnson, and Hefferon 2019) (“How much confidence do you have in scientists to act in the best interests of the public? [1-5; 1 = No confidence at all, 5 = A great deal of confidence]”), the latter having been used in a recent international study on trust in science (Cologna et al. 2024). We selected these items so that we could compare the answers in our sample to global survey results. We find that all three items are highly correlated throughout all studies, and that our results reported here generally replicate when using either of the alternatives measures² (see ESM).

3.3 Results

The main outcome of interest is acceptance of the scientifically consensual facts presented. Overall, acceptance was very high (aggregating across all studies: 95.1 %; Studies 1: 93 %; 2: 98 %; 3: 98 %; 4: 91 %). Note that this includes both participants who had previously correctly answered the knowledge question, and participants who changed their mind when presented with the scientific consensus. In Studies 3 and 4, we gave participants a second chance in case they had initially rejected the consensus, which slightly increased acceptance rates in those studies (initial acceptance in Studies 3: 96 %; 4: 86 %).

As shown in Figure 3.1, these very high rates of acceptance hold for: participants who do not trust science at all (4.2 % of participants, acceptance rate of 87.3 %), participants who rank in the top two deciles of the conspiracy thinking scale (34.3 % of participants, acceptance rate of 93.2 %), participants who consider as “completely true” (the maximum of the 9-point scale) conspiracy theories stating that the earth is flat (3.7 % of participants, acceptance rate of 87.2 %), or that climate change due to fossil emissions is a hoax (11.4 % of participants, acceptance rate of 91.7 %).

Participants in the lowest decile of acceptance still had an average acceptance rate of 67.4 %. Even the three participants who considered as “completely true” that the earth is flat and who said they do “not trust science at all” had an average acceptance rate of 86.7 %.

These high acceptance rates do not merely reflect science knowledge: participants only correctly answered 65.8 % of the questions (Studies 1: 74 %; 2: 79 %; 3: 75 %; 4: 36 %) before they were provided with the scientifically consensual answer. Even the lowest decile in science

²With two exceptions: In Study 3, we find no correlation between acceptance and the Pew question; In Study 4, we find a correlation between knowledge and both alternative trust measures, but not with our main measure (see ESM)

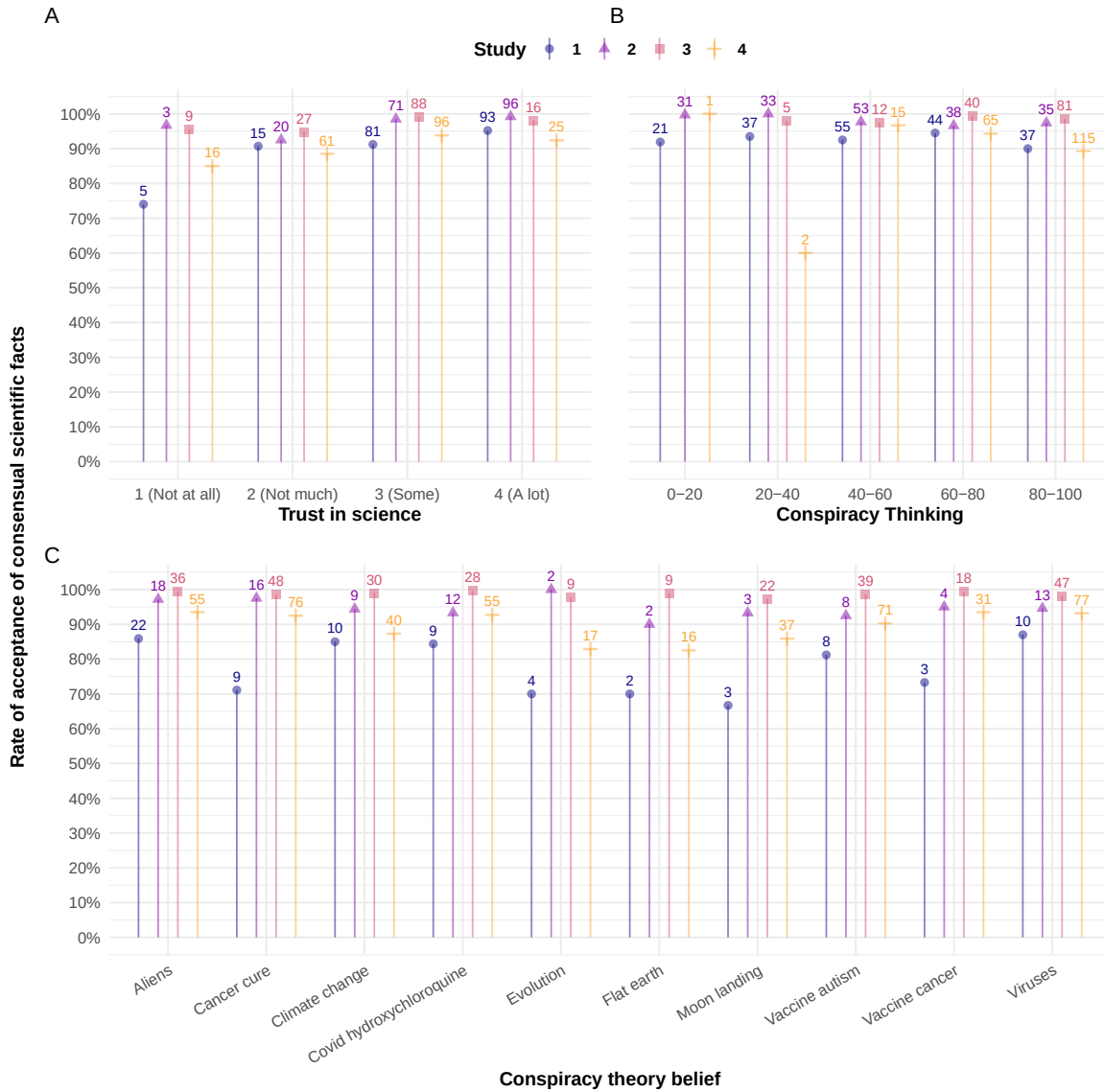


Figure 3.1: Points represent the average share of acceptance and numbers the absolute count of participants as a function of: **A** the level of trust in science (“In general, would you say that you trust science a lot, some, not much, or not at all? [1 = Not at all, 2 = Not much, 3 = Some, 4 = A lot]”); **B** the average conspiracy thinking (CMQ, five items on a scale from 0 to 100); **C** the belief in specific conspiracy theories (i.e. participants who answered 9, “completely true”, for a given theory, see Table 3.2 for the list of the theories).

knowledge, which on average answered correctly only on 20 % of the questions, had an average acceptance rate of 91.7 %.

Did participants who had initially provided a wrong answer change their minds towards the scientific consensus? Yes. In most cases (Studies 1: 76.3 %; 2: 92.9 %; 3: 95.5 %; 4: 89.5 %), participants readily accepted the scientific consensus after having initially given the wrong answer to a question.

How do knowledge and acceptance relate to declared general trust in science and conspiracy belief (two strongly correlated variables, pooled $r = -0.636$, $p < .001$)?

Regarding H1, we find a consistent association between trust in science and acceptance of the scientific consensus (Studies 1: $r = 0.272$, $p < .001$; 2: $r = 0.299$, $p < .001$; 3: $r = 0.198$, $p = 0.019$; 4: $r = 0.155$, $p = 0.029$), but a less consistent relation between trust in science and science knowledge (Studies 1: $r = 0.29$, $p < .001$; 2: $r = 0.279$, $p < .001$; 3: $r = 0.14$, $p = 0.100$; 4: $r = 0.104$, $p = 0.144$).

Regarding H2, the results are mixed for the relation of conspiracy beliefs (measured as the average acceptance of all the conspiracy beliefs) with both acceptance of the scientific consensus (Studies 1: $r = -0.334$, $p < .001$; 2: $r = -0.371$, $p < .001$; 3: $r = -0.023$, $p = 0.788$; 4: $r = -0.048$, $p = 0.498$) and science knowledge (Studies 1: $r = -0.385$, $p < .001$; 2: $r = -0.402$, $p < .001$; 3: $r = -0.164$, $p = 0.055$; 4: $r = -0.023$, $p = 0.742$).

Why did participants reject the scientific consensus? We collected a total of 364 answers (Studies 2: 35; 3: 74; 4: 255) from 167 (Studies 2: 25; 3: 47; 4: 95) participants to the open-ended questions on why they had rejected the scientific consensus on a particular question. Based on the answers, we created five categories (**?@tbl-justifications**). All individual answers can be accessed in cleaned data sheets via the OSF project page.

Table 3.3: Justifications for rejecting the scientific consensus by category, Studies 2, 3 and 4 combined.

Category	N (instances)	Share (instances)	N (participants)*
Not convinced	158	43.4%	70
No justification	103	28.3%	51
Personal convictions	43	11.8%	31
Mistake	42	11.5%	34
Religious Beliefs	18	4.9%	13

Note: *Participants with at least one answer in that category

Why did participants say they accept the scientific consensus? In Studies 3 and 4—the vaccine hesitant samples—we had asked participants about cases in which they agreed with the scientific consensus. A total of 320 (Studies 3: 122; 4: 198) participants answered this question. There were more participants saying they accepted the scientific consensus because they

independently verified the fact (Studies 3: 47.5%; 4: 47%), than participants saying it was because they trust scientists (Studies 3: 41.8%; 4: 36.4%)³. Answers to a question about how they had done so can be found in the ESM.

In an exploratory analysis, we ran linear regressions to test whether there are differences between participants who said they had trusted science and those who said they had verified the information independently. Participants who said they accepted the consensus because of trust in scientists reported trusting science more (Studies 3: mean = 3; $\hat{\beta}_{\text{Trust}} = 0.192$, $p = 0.330$ on a scale from 1 to 4; 4: mean = 2.92; $\hat{\beta}_{\text{Trust}} = 0.465$, $p < .001$) than those who said they verified independently (Studies 3 mean = 2.81; 4: mean = 2.45). We did not find a difference regarding acceptance (Studies 3: $\hat{\beta}_{\text{Acceptance}} = 0.007$, $p = 0.523$ on a scale from 0 to 1; 4: $\hat{\beta}_{\text{Acceptance}} = 0.039$, $p = 0.116$) or regarding beliefs in conspiracy theories (Studies 3: $\hat{\beta}_{\text{BCTI}} = -0.163$, $p = 0.694$ on a scale from 1 to 9; 4: $\hat{\beta}_{\text{BCTI}} = -0.21$, $p = 0.322$). We also did not find a difference in time spent on the survey in Study 3 ($\hat{\beta}_{\text{Time}} = -0.02$, $p = 0.985$; median = 7.6416667 mins), but in Study 4, people who said they had accepted the consensus because they trust scientists tended to spend on average two minutes less on the survey ($\hat{\beta}_{\text{Time}} = -2.159$, $p = 0.018$; median = 9.375 mins).⁴ In Study 4, in which we used facts that participants were unlikely to have encountered before, we tracked whether people clicked on the source links provided—a behavior that you would expect from people who report verifying facts independently. On average, participants clicked only on 1.36 links (out of 20 possible clicks) and there was no difference between the two groups ($\hat{\beta}_{\text{Clicks}} = -1.091$, $p = 0.062$).

More detailed results addressing all our pre-registered research questions can be found in the ESM.

3.4 Discussion

In four studies, we asked US participants whether they accepted scientifically consensual answers on basic science questions. We found quasi-universal acceptance of basic science, with an overall rate of acceptance of 95.1 %, which remained very high for participants who declared not trusting science at all (87.3 % of acceptance), or who endorsed theories blatantly violating scientific knowledge, such as flat earth (87.2 % of acceptance).

This disconnect between declared general trust in science and average trust in (basic) scientific findings goes against predictions from the knowledge–attitudes model of trust in science (see, e.g., Bauer, Allum, and Miller 2007), which posits that knowledge of scientific results is the main cause of trust in science. In line with past research (Allum et al. 2008), we find that both knowledge and acceptance of science are only weakly correlated with general trust in science.

³10.7% in Study 3 and 16.7% in Study 4 answered with “other” and gave an open-ended explanation.

⁴For these analyses, we excluded outliers who took over 30 mins for the survey, which was estimated to take around 10 mins, and for which the median time was seven minutes. As a result, we excluded one participant in Study 3 and four in Study 4. Significance levels are not affected by these exclusions.

This supports recent definitions of science literacy, which suggest that for science literacy to be a meaningful concept for science attitudes, it needs to go beyond measuring knowledge of isolated science facts (National Academies of Sciences, Engineering, and Medicine 2016).

By contrast, our findings align with a motivated reasoning account of trust in science (Lewandowsky and Oberauer 2016), in which “people tend to reject findings that threaten their core beliefs or worldview” (p. 217). A number of participants in our studies endorsed specific conspiracy theories questioning basic tenets of science (on evolution, the shape of the earth, etc.), while still accepting the vast majority of basic scientific facts presented to them. This suggests that these participants had reasons to reject only specific scientific knowledge, and that this rejection prompted them to express a lower trust in science when asked general questions on the topic. Consistent with this explanation, we found a strong association between belief in anti-science conspiracy theories and general trust in science.

Some of the present results also speak to the alienation model (Gauchat 2011), and more specifically to the need for epistemic autonomy (Fricker 2006). Declaring not trusting science, or endorsing conspiracy theories (Harris 2023) might reflect a desire to maintain epistemic autonomy and not appear to ‘blindly’ accept epistemic authority. Such a need to appear epistemically autonomous could be reconciled with the acceptance of basic science facts if it was thought to stem from independent evaluation instead of trust. The majority of participants in Studies 3 and 4, two samples of vaccine-skeptical participants who scored high across a range of different conspiracy beliefs (see ESM), tended to claim that they had accepted the scientific consensus based on their own evaluation, however implausible that might be: it is not clear how participants could independently verify, say, the ratio of glial cells to neurons. These participants did also not engage (more than others) in simple forms of verification, i.e. clicking on sources provided in the answers. This result aligns with other results on conspiracy thinking: Recent studies have shown that, although they claim not to be, conspiracy theorists are in fact just as susceptible to social influence as others (Pummerer et al. 2024; Altay et al. 2023).

In applied terms, the present results have implications for science communication. Across various domains of science knowledge such as climate change (Večkalov et al. 2024) or vaccination (Salmon et al. 2015), researchers have observed a consensus gap: a gap between the scientific consensus and public opinion. Some of the most worrying consensus gaps relate to climate change, as substantial segments of the population disagree with scientists on what is happening and what to do about it (e.g., Egan and Mullin 2017). Yet, we found that for much of basic, non-contentious science, such consensus gaps are absent: The vast majority of our participants—including two vaccine-skeptical samples (Studies 3 and 4)—did not appear to have general grounds for distrusting science, which should have led them to reject most or all of the science knowledge presented to them. Since even people who say they distrust science, or who reject specific scientific facts appear to accept most of basic science, stressing the basic science components of publicly controversial fields, from GMOs to climate change, might help reduce the consensus gaps observed in these domains (on climate change, see, e.g. Ranney and Clark 2016).

The present studies have a number of limitations, in particular the lack of representative samples, and the focus on a single country. While our results support motivated reasoning accounts of trust in science, they leave important questions unaddressed, in particular: Why do people reject specific scientific findings but not others? Suggestions have already been made for a number of issues such as vaccination (Miton and Mercier 2015), GMOs (Blanke et al. 2015), or nuclear energy (Hacquín et al. 2021). However, research is still needed to better understand what motivates these rejections (see e.g., Hornsey 2020). Moreover, rejection of scientific facts can manifest in various ways, for example calling into question the integrity of scientists (Mede 2023; Mede et al. 2022), or denying the very possibility of scientifically investigating certain issues (Munro 2010). The classification of justifications we present here does not address these processes in detail. Finally, by design, most of the basic science knowledge presented in this study didn't directly relate to anything controversial for most people. Future studies could look at basic science that does relate to scientific topics which are the object of public and policy-relevant debates (e.g. GM food, climate science, vaccines), or scientific findings for which there is a more obvious potential conflict of interest (e.g. non-publicly funded medical research).

3.4.1 Data availability

3.4.2 Code availability

The code used to create all results (including tables and figures) of this manuscript is also available on the OSF project page (<https://osf.io/8utsj/>).

3.4.3 Competing interest

The authors declare having no competing interests.

4 Trusting but forgetting impressive science

Trust in science is associated with significant outcomes such as intent to vaccinate and belief in climate change. Around the globe, most people trust science at least to some extent. However, the causes of this trust aren't well understood. Here we propose a 'rational impression' model of trust in science. In this model, people trust scientists because they are impressed by their findings, and this impression of trust persists even after knowledge of the specific content has vanished. We present evidence for this model in two experiments (total $n = 696$) with UK participants. In Experiment 1, impressive scientific findings lead participants to think of the scientists as more competent and their scientific discipline as more trustworthy. In Experiment 2, we show that participants have these impressions despite forgetting the content that generated them. The rational impression model can explain why people trust science without remembering much of it. It also stresses the relevance of science communication.

4.1 Introduction

Jon D. Miller, a pioneer in measuring science literacy, found that at the end of the 20th century, approximately 17% of Americans qualified as scientifically literate—a measure he described as the “level of skill required to read most of the articles in the Tuesday science section of The New York Times” (Miller 2004). His estimate for the early 1990s had been even lower, around 10%. By contrast, the level of trust in science in the US has been relatively high and remarkably stable (Funk and Kennedy 2020; Funk et al. 2020; T. W. Smith and Son 2013): Since the 1970s, on average 40% of Americans say they have a great deal of confidence in the scientific community, the second highest score among 13 institutions included in the US General Social Survey (GSS). How can we explain this gap between knowing science and trusting it? Here, we argue that people might trust science in large part because they have been impressed by it.

We begin by presenting global evidence that most people trust science at least to some extent. We then review existing explanations for why people trust science, as well as empirical observations that seem to violate their predictions. As an alternative explanation, we propose the 'rational impression model'. Two experiments provide evidence in favor of this model.

Trust in science matters. It is related to many desirable outcomes, from acceptance of anthropogenic climate change (Cologna and Siegrist 2020) or vaccination (Sturgis, Brunton-Smith, and Jackson 2021; Lindholt et al. 2021) to following recommendations during COVID (Algan et al. 2021).

Across the globe, most people report trusting science at least to some extent (Wellcome Global Monitor 2018, 2020). Recently, a large survey in 68 countries found trust in scientists to be “moderately high” across all countries (mean = 3.62; sd= 0.70; Scale: 1 = very low, 2 = somewhat low, 3 = neither high nor low, 4 = somewhat high, 5 = very high), with not a single country below midpoint trust. A recent study in the US found that participants almost always accepted the scientific consensus on basic, non-politicized knowledge questions (e.g. ‘Are electrons are smaller than atoms?’), even those participants who said they do not trust science and who believed anti-science conspiracy theories, e.g. that the earth is flat (Pfänder, Kerzreho, and Mercier 2024).

How can we explain trust in science? The literature on the public understanding of science has generally focused not on explaining trust, but distrust, and more generally negative attitudes towards science (Bauer, Allum, and Miller 2007).

One explanation for distrust towards science is the *alienation model* (Gauchat 2011). According to this model, the “public disassociation with science is a symptom of a general disenchantment with late modernity, mainly, the limitations associated with codified expertise, rational bureaucracy, and institutional authority” (Gauchat 2011, 2). This explanation builds on the work of social theorists (Habermas 1989; Beck 1992; Giddens 1991; see Gauchat 2011 for an overview) who suggested that a modern, complex world increasingly requires expertise, and thus shapes institutions of knowledge elites. People who are not part of these institutions experience a lack of agency, resulting in a feeling of alienation. If the alienation model might account for some of the distrust towards science, its goal is not to explain why most people still trust science.

A second explanation which has dominated the literature on public understanding of science for decades (Bauer, Allum, and Miller 2007), is the deficit model. According to the deficit model, people do not trust science enough, because they do not know enough about it. The deficit model implies that when people trust science, it is because they are knowledgeable about it (Bauer, Allum, and Miller 2007).

In line with that view, large scale survey studies consistently identify education, and in particular science education, to be the strongest correlate of trust in science (Bak 2001; Noy and O’Brien 2019; Wellcome Global Monitor 2018, 2020): More educated people tend to trust science more. The deficit model provides a straightforward causal explanation for this correlation: education fosters trust in science by transmitting knowledge and improving understanding of science.

Evidence for the role of knowledge in trust in science, however, is far from conclusive. While there is no consensus among researchers on how to measure understanding of science (Gauchat and Andrews 2018), a widely used measure consists of asking people a set of basic science

knowledge questions (see e.g. Durant, Evans, and Thomas 1989; Miller 1998). Past research has found science knowledge to be alarmingly low (Miller 2004; National Academies of Sciences, Engineering, and Medicine 2016), contrasting with the relatively high levels of trust in science. Moreover, science knowledge is only weakly associated with attitudes towards science (Allum et al. 2008). Other research has shown that the correlation between education and positive attitudes towards science holds even after controlling for science knowledge, which led the researchers to conclude that only part of the effect of education could be explained by transmitting enduring knowledge of science (Bak 2001).

4.2 The rational impression model

Here, we seek to explain the following stylized facts: (i) Globally, most people have at least some trust in science, and essentially everyone, at least in the US, trusts nearly all of basic science; (ii) Education, and in particular science education, is the strongest predictor of trust in science; (iii) Levels of science knowledge are relatively low, and the association between science knowledge and trust in science is weak.

All of these observations are compatible with a model of trust in science which we dub the *rational impression model*. According to this model, people trust science because they are impressed by it, but they forget about the findings that impressed them.

For an information to be impressive, at least two criteria should be met: (i) it is perceived as hard to uncover and (ii) there is reason to believe it true. By (i), we mean information that most people would have no idea how to find out themselves. For example, most people would probably only be mildly impressed by someone telling them that a given tree has exactly 110,201 leaves. Even though obtaining this information implies an exhausting counting effort, everyone in principle knows how to do it. By contrast, finding out that it takes light [approximately 100,000 years to travel from one end of the Milky Way to the other](#) is probably impressive to most people, as they would not know how such a distance can be measured. Regarding (ii), people can sometimes judge the accuracy of scientific findings for themselves, for example when they are exposed to accessible and convincing explanations in school, e.g. simple explanations that have a broad explanatory scope (Read and Marcus-Newhall 1993; Lombrozo 2007; for a review, see Lombrozo 2006). When people cannot evaluate the quality of the information for themselves, they need to rely on other cues to establish whether it might be true or not. One such cue is the degree of consensus between different sources. It has been shown that when they lack relevant background knowledge, people rely on social consensus as a heuristic to judge the accuracy of a piece of information and the competence of its source (Pfänder, De Courson, and Mercier 2025). This heuristic is rational, in the sense that it leads to sound inferences if the sources are independent and unbiased. As a result, unless we suspect a conspiracy among scientists, we should be impressed when they agree on something like the length of the Milky Way, and think of the scientists as competent, even if we have no idea how they got to agree on this measure.

Once people are impressed, that impression tends to stick, even if specific content is forgotten. We commonly form impressions of the people around us while forgetting the details of how we formed these impressions: If a colleague fixes our computer, we might forget exactly how they fixed it, yet remember that they are good at fixing computers. As an extreme example, patients with severe amnesia can continue to experience emotions linked to events they could not recall (Feinstein, Duff, and Tranel 2010). More relevantly here, a study has shown that while people find some science-related explanations more satisfying than others, this did not predict how well they could recall the explanations shortly after (Liquin and Lombrozo 2022), suggesting that that impressions and knowledge formation can be quite detached.

The rational impression model of trust in science reconciles the stylized facts outlined above: If most people trust science (i), it is because most people receive at least some basic education that exposes them to impressive scientific findings, setting up a baseline of trust in science. If education is the main predictor of trust in science (ii), it is because education is the primary mean through which people are exposed to science. If the levels of science knowledge are low while trust in science is high (iii), it is because specific knowledge of science is forgotten, while the impression of competence and trust is maintained.

While the rational impression model, we have argued, can reconcile well-established facts about trust in science and science knowledge, there is no experimental evidence directly testing its central mechanism. This is the goal of the present studies.

4.3 The present studies

We tested the predictions of the rational impression model in two experiments (total $n = 696$): Experiment 1 tests whether impressive scientific findings lead people to think of the scientists as more competent and more trustworthy. Experiment 2 tests whether that impression persists even if specific content is forgotten. Figure 4.1 provides an overview of the main results of the two experiments.

Both experiments were preregistered, and the choice of sample size was informed by power simulations. All materials, data, and code can be found on Open Science Framework [project page](https://osf.io/j3bk4/) (<https://osf.io/j3bk4/>). All analyses were conducted in R (version 4.2.2) using R Studio. We used two-sided tests for all hypotheses. Unless mentioned otherwise, we report unstandardized estimates that can be interpreted in units of the original scales.

As part of this project, we conducted two additional experiments which we present in detail in the ESM. The first experiment (‘Experiment 1b’ in the ESM) is almost identical to Experiment 1, but suffered from a minor technical error during the implementation. Nonetheless, its findings are identical to those of Experiment 1. The second experiment (‘Experiment 2b’) was supposed to test the same hypothesis as Experiment 2, but the treatment—a very short distraction task—did not alter any of the outcome variables, suggesting that our outcome measures, a set of multiple-choice questions, was not sufficiently sensitive.

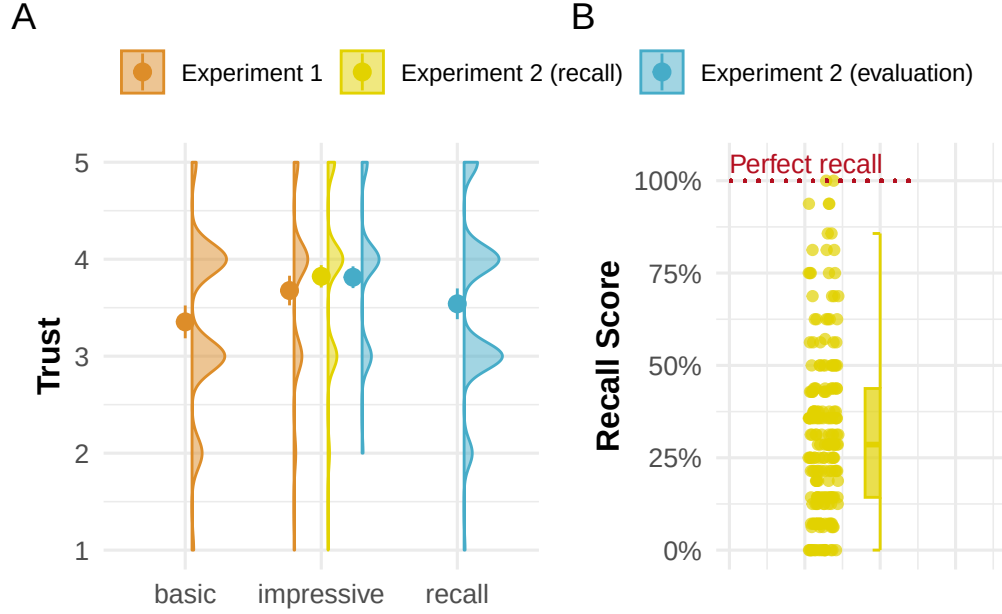


Figure 4.1: **A.** An overview of the differences in trust, according to whether the participants had read a basic, an impressive, or another participant’s recalled version of the impressive vignette. The density plots show the distributions of participants’ trust scores. The dots represent model estimates, and the vertical bars the 95% confidence intervals of the model predictions. **B.** The distribution of knowledge scores (ranging from 0% to 100% retained information) in the recall study of Experiment 2. Each dot corresponds to one participant. For the boxplot, the box represents the interquartile range (IQR), that is, the distance between the first and third quartiles, the center line indicates the median, and the outer lines (whiskers) extend to 1.5 times the IQR or the most extreme values within this range.

4.4 Experiment 1

The goal of Experiment 1 was to test whether exposure to impressive science content enhanced people’s trust in scientists and their discipline. Participants were presented with vignettes about scientific findings in the disciplines of entomology and archaeology. The impressiveness of the texts was manipulated by creating one ‘basic’ and one ‘impressive’ version for each of the disciplines (see Table 4.1). Impressiveness was manipulated within participants, but between disciplines: each participant was randomly assigned to see an impressive version for one discipline, and a basic version for the other discipline. We tested the following hypotheses:

H1a: After having read an impressive text about a discipline’s findings, compared to when reading a basic text, participants perceive that discipline’s scientists as more competent.

H1b: Across both conditions, participants who are more impressed by the text about a discipline also tend to perceive the scientists of that discipline as more competent.

H2a: After reading an impressive text about a discipline’s findings, compared to when reading a basic text, participants will trust the discipline more.

H2b: Across both conditions, participants who are more impressed by the text about a discipline also tend to trust the discipline more.

The results of two research questions, about perceptions of learning and of consensus, are reported in the ESM.

4.4.1 Methods

4.4.1.1 Participants

A power simulation (see OSF) suggested that the minimum required sample size to detect a statistically significant effect for all hypotheses with a power of 0.9 is 100 participants. We therefore recruited 100 participants from the UK via Prolific. One participant failed the attention check, resulting in a final sample of 99 participants (49 female, 50 male; age_{mean} : 42.071, age_{sd} : 12.857, age_{median} : 40).

4.4.1.2 Procedure

After providing their consent to participate in the study and passing an attention check (see ESM), participants read a short introductory text, and then two vignettes (one basic and one impressive) about scientific findings in the disciplines of entomology and archaeology, in a randomized order. After reading each vignette, participants were asked: “How much do you feel you’ve learnt about [human history/insects] by reading this text?” [1 - Nothing, 2 - A bit, 3 - Some, 4 - Quite a bit, 5 - A lot]; “How impressive do you think the findings of

the [archaeologists/entomologists] described in the text are?” [1 - Not very impressive, 2 - A bit impressive, 3 - Quite impressive, 4 - Very impressive, 5 - Extremely impressive]); “Would you agree that reading this text has made you think of [archaeologists/entomologists] as more competent than you thought before?” [1 - Strongly disagree, 2 - Disagree, 3 - Neither agree nor disagree, 4 - Agree, 5 - Strongly agree]); and “Having read this text, would you agree that you trust the discipline of [archaeology/entomology] more than you did before?” [1 - Strongly disagree, 2 - Disagree, 3 - Neither agree nor disagree, 4 - Agree, 5 - Strongly agree]. Finally, we asked: “To which extent do you think the findings from the short text you just read reflect a minority or a majority opinion among archaeologists?” [1 - Small minority, 2 - Minority, 3 - About half, 4 - Majority, 5 - Large majority].

4.4.1.3 Materials

Table 4.1 shows the stimuli used in Experiment 1.

Table 4.1: Stimuli of Experiment 1

	Impressive	Basic
Archeology	Archaeologists, scientists who study human history and prehistory, are able to tell, from their bones, whether someone was male or female, how old they were, and whether they suffered from a range of diseases. Archaeologists can now tell at what age someone, dead for tens of thousands of years, stopped drinking their mother’s milk, from the composition of their teeth. Archaeologists learn about the language that our ancestors or cousins might have had. For instance, the nerve that is used to control breathing is larger in humans than in apes, plausibly because we need more fine-grained control of our breathing in order to speak. As a result, the canal containing that nerve is larger in humans than in apes – and it is also enlarged in Neanderthals. Archaeologists can also tell, from an analysis of the tools they made, that most Neanderthals were right-handed. It’s thought that handedness is related to the evolution of language, another piece of evidence suggesting that Neanderthals likely possessed a form of language.	Archaeology is the science that studies human history and prehistory based on the analysis of objects from the past such as human bones, engravings, constructions, and various objects, from nails to bits of pots. This task requires a great deal of carefulness, because objects from the past need to often be dug out from the ground and patiently cleaned, without destroying them in the process. Archaeologists have been able to shed light on human history in all continents, from ancient Egypt to the Incas in Peru or the Khmers in Cambodia. Archaeologists study the paintings made by our ancestors, such as those that can be found in Lascaux, a set of caves found in the south of France that have been decorated by people at least 30000 years ago. Archaeologists have also found remains of our more distant ancestors, showing that our species is just one among several that appeared, and then either changed or went extinct, such as Neanderthals, Homo erectus, or Homo habilis.

Table 4.1: Stimuli of Experiment 1 (*continued*)

	Impressive	Basic
Entomology	Entomologists are the scientists who study insects. Some of them have specialized in understanding how insects perceive the world around them, and they have uncovered remarkable abilities. Entomologists interested in how flies' visual perception works have used special displays to present images for much less than the blink of an eye, electrodes to record how individual cells in the flies' brain react, and ultra-precise electron microscopy to examine their eyes. Thanks to these techniques, they have shown that some flies can perceive images that are displayed for just three milliseconds (a thousandth of a second) – about ten times shorter than a single movie frame (of which there are 24 per second). Entomologists who study the hair of crickets have shown that these microscopic hairs, which can be found on antenna-like organs attached to the crickets' rear, are maybe the most sensitive organs in the animal kingdom. The researchers used extremely precise techniques to measure how the hair reacts to stimuli, such as laser-Doppler velocimetry, a technique capable of detecting the most minute of movements. They were able to show that the hair could react to changes in the motion of the air that had less energy than one particle of light, a single photon.	Entomologists are scientists who investigate insects, typically having a background in biology. They study, for example, how a swarm of bees organizes, or how ants communicate with each other. They also study how different insects interact with each other and their environment, whether some species are in danger of going extinct, or whether others are invasive species that need to be controlled. Sometimes entomologists study insects by observing them in the wild, sometimes they conduct controlled experiments in laboratories, to see for example how different environmental factors change the behavior of insects, or to track exactly the same insects over a longer period of time. An entomologist often specializes in one type of insect in order to study it in depth. For example, an entomologist who specializes in ants is called a myrmecologist.

4.4.2 Results and discussion

As a manipulation check, we find that participants perceived the impressive texts to be more impressive (mean = 4.01, sd = 0.87; $\hat{b} = 0.808$ [0.569, 1.048], $p < .001$) than the basic texts (mean = 3.2, sd = 1.13).

Participants perceived scientists as more competent after having read an impressive text (H1a: $\hat{b}_{\text{Competence}} = 0.495$ [0.314, 0.676], $p < .001$; mean = 3.76, sd = 0.9) than after having read a basic one (mean = 3.26, sd = 0.8). Pooled across both conditions, participants' impressiveness ratings were positively associated with competence: When participants reported being more impressed, they evaluated scientist#s as more competent (H1b: $\hat{b} = 0.407$ [0.312, 0.502], $p < .001$).

Participants also trusted a discipline more after having read an impressive text (H2a: $\hat{b}_{\text{trust}} = 0.323$ [0.17, 0.476], $p < .001$; mean = 3.68, sd = 0.87) than after having read a basic one. Participants' impressiveness ratings were positively associated with trust when pooling across all conditions (H2b: $\hat{b} = 0.296$ [0.206, 0.385], $p < .001$).

4.5 Experiment 2

In Experiment 1, exposure to impressive scientific content increased trust in the relevant scientific discipline, and the perceived competence of the relevant scientists. Experiment 2 sought to test whether these perceptions were at least partly independent of being able to recall the specific content that had induced them. Experiment 2 consisted of a ‘recall study’ and an ‘evaluation study.’ In the recall study, each participant was assigned to read the impressive version of one of the vignettes from Experiment 1 (Table 4.1). As in Experiment 1, participants were asked about how impressed they were by the findings and whether they had changed their perception of the scientists’ competence and their trust in the scientific discipline.

In line with the findings of Experiment 1, we expected participants to have increased perceptions of competence and trust in scientists after having read the impressive texts:

H1a: Participants perceive scientists as more competent after having read an impressive text about their discipline’s findings.

H1b: Participants trust a discipline more after having read an impressive text about the discipline’s findings.

In Experiment 2, participants were also given a recall task: They were asked to rewrite the text of the vignette they had just read, from memory. We predicted that participants would not be able to recall all the information presented in the short vignettes right after having read them (see methods section for details):

H2: Participants are not able to recall all the information of the original texts.

This hypothesis, however, only tested whether people forgot any of the content, potentially including non-impressive content. To address this issue, we asked participants to select elements of the vignettes they found impressive (see Table 4.2). We predicted that even for the subset of information that a participant said they found impressive, they forgot at least some of the content:

H3: Participants are not able to recall all the impressive information—as rated by themselves—contained in the original text.

H3 tests the hypothesis that participants immediately forget at least some of the information that has impressed them. However, it could be that participants in fact remember enough impressive information to justify the increase in perceived trust (in the discipline) and competence (in the scientists). To test whether that was the case, we conducted an evaluation study.

A new sample of participants was recruited, and they were randomly assigned to one of two conditions: In the ‘original impressive text’ condition, participants were assigned to read one of the two impressive texts of the recall study. In the ‘recalled impressive text’ condition, participants read one of the recall texts written by the participants of the recall study. We

predicted that participants in the evaluation study would be less impressed by the texts recalled by the participants of the recall study, compared to the original impressive vignette texts (see Table 4.1) and, accordingly, would have less positive perceptions of the scientists' competence and the trustworthiness of their discipline:

H4a: The texts produced by participants of the experiment as a result of the recall task will be less impressive than the original texts, as rated by participants of the **evaluation study**.

H4b: Participants of the **evaluation study** perceive scientists as more competent after having read the original texts, compared to after having read the texts produced by participants of the experiment as a result of the recall task.

H4c: Participants of the **evaluation study** trust a discipline more after having read the original texts, compared to after having read the texts produced by participants of the experiment as a result of the recall task.

4.5.1 Methods

4.5.1.1 Participants

In a power simulation (see OSF), we varied the sample size of the recall study and effect sizes (assuming the same effect size for all hypotheses in each scenario). For all simulations, a constant evaluation study sample size of 400 participants was assumed (only relevant for H4a, b and c). The power simulation suggested that a power level of 90% would be reached when assuming a medium effect size of 0.5 with 50 participants. Due to uncertainty about our assumptions, we recruited a sample of 203 participants for the recall study, and a sample of 406 participants for the evaluation study. Although this was not the focus of the simulation, the results showed that for a medium effect size of 0.5, a sample size of 400 for the evaluation study yielded statistical power of greater than 90% for all hypotheses based on this sample (H4a, b and c). All participants were from the UK, recruited via Prolific, and paid to complete the experiment.

The final sample comprised 198 participants (four failed attention checks; 99 female, 99 male; age_{mean} : 42.646, age_{sd} : 15.547, age_{median} : 40) for the recall study, and 399 participants for the evaluation study (seven failed attention checks; 201 female, 198 male; age_{mean} : 41.709, age_{sd} : 13.483, age_{median} : 40).

4.5.1.2 Procedure

4.5.1.2.1 Recall study

In the recall study, after having consented to take part in the study and passing an attention check (see ESM), participants read the impressive version of one of the vignettes from Experiment 1 (Table 4.1). After reading the text, as in Experiment 1, participants were asked

about changes in their perception of the scientists’ competence, and trust in the scientists’ discipline. They were also asked about the impressiveness of the text they read. The order of these questions was randomized.

Next, as an open-ended question, participants were asked to recall as much information as they could of the texts they had just read. They were told that their texts would be read by future participants. To further motivate participants, they were also told that they would get a bonus for recalling (without external aids) accurate information. They were not told how much that bonus would be. We paid them 5p per point gained in the recall task (see methods for how these points were assigned). This way, participants could reach a maximum bonus of 0.8 pound for archaeology (0.05p x 2 points x 8 content elements) and 0.7 pound (0.05p x 2 points x 7 content elements) for entomology. After that, participants were presented with the evaluation grid that we used to assess the open answers from the recall task (see Table 4.2). For each knowledge element, we asked participants to indicate whether they found it impressive or not (“Do you think this piece of information is impressive?” [Yes; No]). At the end of the recall study, participants were asked about their education level.

4.5.1.2.2 Evaluation study

After consenting to taking part in the study and passing an attention check (see ESM), participants read either one of the original vignettes, or a text produced as part of the recall task from the recall study. For those participants assigned to read a recall text, the text was randomly sampled (with replacement) from a pool of recall answers. Orthographic and grammatical mistakes in these texts were corrected with the help of ChatGPT beforehand. We had preregistered to sample among all recall answers from participants of the recall study. However, checking recall scores after the recall study and before launching the evaluation study, we found them to be very low on average (see ESM section @ref(above-median)). To avoid having the answers of less motivated participants in our sample, we decided to only select answers that scored at least as well as the median in the recall measure of the evaluation study. This selection is conservative in that it makes it harder to confirm our predictions under H4.

After reading the text, just as in the recall study, participants were asked about changes in their perception of the scientists’ competence, trust in the scientists’ discipline, and the impressiveness of the text they read. The order of these questions was randomized. Finally, participants were asked about their education level.

4.5.1.3 Materials

For Experiment 2, besides the texts recalled by the participants of the recall study, the impressive version of the stimuli used in Experiment 1 was used (see Table 4.1).

Table 4.2: Recall evaluation grid

Archaeology	Entomology
1. Archaeologists can determine whether someone was male or female from their bones.	1. Entomologists use special displays to present images to flies for extremely short periods (less than the blink of an eye).
2. Archaeologists can determine how old someone was from their bones.	2. Entomologists can record how individual cells in flies' brains react using electrodes.
3. Archaeologists can determine whether someone suffered from a range of diseases from their bones.	3. Entomologists use ultra-precise electron microscopy to examine flies' eyes.
4. Archaeologists can determine at what age someone stopped drinking their mother's milk, based on the composition of their teeth.	4. Some flies can perceive images displayed for just three milliseconds. This duration is about ten times shorter than a single movie frame.
5. The nerve controlling breathing is larger in humans than in apes. The canal containing that nerve is also larger in humans and Neanderthals than in apes.	5. Crickets have microscopic hairs situated on antenna-like organs at their rear.
6. The fact that the nerve controlling breathing is larger in humans is possibly due to the need for fine-grained control of breathing to speak.	6. Crickets' hairs are possibly the most sensitive organs in the animal kingdom. They react to changes in air motion with less energy than one photon.
7. Archaeologists determined that most Neanderthals were right-handed, based on analysis of Neanderthals' tools.	7. Entomologists measured how cricket hairs react to stimuli, using laser-Doppler velocimetry, which can detect extremely minute movements.
8. Handedness is thought to be related to the evolution of language. This suggests that Neanderthals likely possessed a form of language.	

Note:

For each knowledge element, participants could score a maximum of two points.

4.5.1.3.1 Recall

As shown in Table 4.2, the texts were divided into a series of knowledge elements. For each participant, the extent to which they recalled each of the different elements was coded. A recall score based on how many of these elements they mentioned in their open-ended answer was then calculated.

The coding was done with the help of ChatGPT (see ESM section @ref(gpt-prompt) for the exact prompt). The instructions were: to code 0 if a piece of knowledge was not mentioned or was mentioned with significant errors (e.g., writing “the nerve controlling fine hand movement

is bigger in humans” instead of “the nerve controlling breathing is bigger in humans”); to code 1 if the piece of knowledge was mentioned, but some important elements were missing (e.g., writing “Archaeologists can determine at what age someone stopped drinking their mother’s milk” instead of “Archaeologists can determine at what age someone stopped drinking their mother’s milk from the composition of their teeth”), and/or there were some mistakes (e.g., writing “Archaeologists can determine at what age someone stopped drinking their mother’s milk, based on the bones” instead of “Archaeologists can determine at what age someone stopped drinking their mother’s milk based on the teeth”); to code 2 if the piece of knowledge was mentioned with all the main content, even if the participant had not used the precise technical words (e.g., “neanderthals,” “laser-Doppler velocimetry”) or had changed the phrasing in other ways. These instructions were intended to produce relatively generous recall scores.

Since the two vignettes contained a different number of total knowledge elements according to our evaluation grid (8 for archaeology, 7 for entomology), we used a relative measure for the final recall score, namely the share of obtained points among all possible points (possible range from 0 to 1, below the results are presented in percentages recalled for clarity). Practically, for all tests on recall, we computed a forgetting score (1-recall score), and tested whether it was statistically significantly different from zero.

To validate the scores assigned by ChatGPT, they were compared to scores assigned by two human coders for a subsample of 80 randomly chosen texts (half on archaeology, half on entomology). The human coders were unaware of the study context and the hypotheses. They were provided with the prompt given to ChatGPT. To measure the agreement between ChatGPT and the human coders, we calculated an intraclass correlation coefficient (ICC) for two-way designs with random raters (Ten Hove, Jorgensen, and Ark 2024). Following the guidelines in Heyman et al. (2014), we preregistered taking 0.7 as a threshold for acceptable reliability. In our sample, we observe an ICC of 0.838 [se = 0.032]. Human coders and ChatGPT, on average, assigned exactly the same score in 72.9% of all rating instances (compared to 73.5% of exact agreement between the two human coders). Following our preregistration, we consider the high agreement as suggested by the ICC a validation of the use of ChatGPT.

4.5.1.3.2 Recall of impressive items

After giving their post-recall evaluation of scientists’ competence and trust in the discipline, participants were presented with the evaluation grid shown in Table 4.2 for the respective discipline they had been randomized to see. For each element in the evaluation grid, they were asked whether they found it impressive or not. Then, for each participant, the recall score was computed just as described above, but only on the subset of those elements they had subjectively rated as impressive.

4.5.2 Results and discussion

4.5.2.1 Recall study

First, as a kind of validation check, we note that most participants declared finding all knowledge elements to be impressive ($mean_{Archeology} = 7.04$, $median_{Archeology} = 8$, number of knowledge elements = 8; $mean_{Entomology} = 6.39$, $median_{Entomology} = 7$, number of knowledge elements = 7).

For the first set of hypotheses, we tested whether the average scores for perceived change in competence and trust were significantly different from their respective scale midpoints (which corresponds to no change in perception). We first tested the outcome variables' distributions for normality, using a Shapiro-Wilk test. In all cases, this test suggested that the data is considered non-normally distributed ($p < 0.05$). Following the preregistration, we therefore did not run a default one-sample t-test, but used a Wilcoxon signed-rank test instead. H1a and H1b were both supported: After having read an impressive text about the findings of a scientific discipline, participants saw the scientists as more competent (H1a: median = 4, $W = 12388.5$, $p < .001$), and their discipline as more trustworthy (H1b: median = 4, $W = 9714.5$, $p < .001$).

Despite having been impressed, a first descriptive analysis suggested that participants seemed to recall only very little information ($mean = 32\%$, $sd = 23\%$, $median = 29\%$). Given these low average scores, we opted for a more conservative approach to testing H2 and H3: We defined the median as a cut-off and selected only the 50% of participants who had the highest recall scores, removing participants who may have put in less effort. Since we hypothesized that participants would forget content, this selection made it less likely that the hypotheses would be confirmed. Even for the 50% participants with the best recall, both hypotheses were supported: Participants did not perfectly recall all information (H2: median = 0.43, $W = 5778$, $p < .001$) and they did not recall all information contained in the elements they judged as impressive themselves (H3: median = 0.43, $W = 5778$, $p < .001$)¹.

4.5.2.2 Evaluation study

For H4a, b and c, trust, competence and impressiveness ratings were compared between the 'original impressive text' and the 'recalled impressive text' conditions of the evaluation study using independent sample t-tests. Participants who read one of the two original impressive texts reported being more impressed (H4a: $\hat{b}_{Impressiveness} = 0.387$, $t = 5.411$, $p < .001$), rated the scientists of the respective discipline as more competent (H4b: $\hat{b}_{Competence} = 0.274$, $t = 3.074$, $p = 0.002$), and had more trust in the respective discipline (H4c: $\hat{b}_{Trust} = 0.274$, $t = 3.398$, $p < .001$) than participants who read one of the texts of the recall task.

¹The two median values for all information and for the subset of impressive information are not exactly the same, but very similar, because participants rated most knowledge elements as impressive, making the general recall score and the recall score for impressive elements very similar.

4.6 Discussion

It has long been a puzzle to the deficit model—which suggests that trust in science is primarily driven by science knowledge—that knowledge of science is at best weakly associated with science attitudes (Allum et al. 2008; National Academies of Sciences, Engineering, and Medicine 2016). The rational impression model makes sense of this: It predicts that people trust science primarily because they have been impressed by it, not because they remember much of the knowledge that impressed them.

Two experiments provide evidence for this model. Experiment 1 showed that impressive scientific findings lead people to think of scientists as more competent and trust science more. Experiment 2 showed that these impressions are formed even though participants forget the content that generated them almost immediately after reading it.

How can this model be rational, if it posits that trust is largely detached from knowledge? It is rational in that the heuristics it builds on lead to sound inferences in many contexts. If someone discovers something that is hard to know, such as the size of the Milky Way, and there appears to be a consensus, we should expect them to be competent, even without knowing the details of how they made this discovery. Even forgetting specific knowledge is not irrational: It has been argued that one of the main functions of episodic memory is to justify our beliefs in communication with others (Mahr and Csibra 2018). As a result, we should be particularly good at remembering things we might need to convince others of. In this regard, incentives of remembering science seem to be weak: Most exposure to science happens at school, and there is little reason for young learners’ minds to anticipate having to convince others of the merits of specific scientific findings, which are typically of little practical relevance to them, and which appear to be consensually accepted.

Does this mean that scientific education should focus on impressive but potentially hard to grasp findings, instead of fostering a proper understanding of more basic discoveries? No, for at least two reasons. First, for some students at least, this knowledge will be remembered, and will prove important in their lives. Second, positive impressions of science can also be created by a proper understanding of its explanations, methods, etc. – even if, once again, that understanding is then often lost. However, our findings do suggest that science educators should not despair when they observe the low rates of science knowledge in adults, since the exposure to science they provided is arguably responsible for creating a bedrock of trust in science.

The present framework also stresses the vital role that can be played by science communication. The relationship between science communication and trust in science has already been explored in depth (e.g., Weingart and Guenther 2016; for a recent review see König et al. 2023), but we believe the present model might make a useful contribution. In particular, it suggests that, even though more understanding of the underlying methods is always preferable (König et al. 2023), even a relatively superficial exposure to impressive findings can bolster trust in science. An important caveat is that impressive findings that haven’t yet gained the approval of the

community might be particularly likely to backfire if people learn they have been disproven (on the importance of presenting a measure of consensus alongside scientific information, see König et al. 2023).

The present experiments have a number of limitations: First, they were conducted on convenience samples recruited in a single country, the UK. Second, they were conducted within a very short time frame. While we can show that participants almost immediately forget about impressive content, it is not clear from our study for how long the impressions persist (although in other contexts impressions formed on the basis of a much more superficial exposure have been shown to last for months, Gunaydin, Selcuk, and Zayas 2017). Future studies could extend our findings to other populations and to longer time frames.

4.6.0.1 Data availability

Data for all experiments and the simulations is available on the OSF project page (<https://osf.io/j3bk4/>). Note that on the project page, all materials related to what is referred to as “Experiment 1” in this paper are stored under “experiment_2”, and all materials related to “Experiment 2” in this paper are stored under “experiment_4”. This numbering is due to the original order in which experiments for this project were conducted. For a detailed report on the other experiments conducted as part of this project, see the ESM.

4.6.0.2 Code availability

The code used to create all results (including tables and figures) of this manuscript is also available on the OSF project page (<https://osf.io/j3bk4/>).

4.6.0.3 Competing interest

The authors declare having no competing interests.

5 How wise is the crowd: Can we infer people are accurate and competent merely because they agree with each other?

Are people who agree on something more likely to be right and competent? Evidence suggests that people tend to make this inference. However, standard wisdom of crowds approaches only provide limited normative grounds. Using simulations and analytical arguments, we argue that when individuals make independent and unbiased estimates, under a wide range of parameters, individuals whose answers converge with each other tend to have more accurate answers and to be more competent. In 6 experiments (UK participants, total $N = 1197$), we show that participants infer that informants who agree have more accurate answers and are more competent, even when they have no priors, and that these inferences are weakened when the informants were systematically biased. In conclusion, we speculate that inferences from convergence to accuracy and competence might help explain why people deem scientists competent, even if they have little understanding of science.

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For supplementary materials, please refer either to the published version, or [the publicly available preprint via github](#).

5.1 Introduction

Imagine that you live in ancient Greece, and a fellow called Eratostenes claims the circumference of the earth is 252000 stades (approximately 40000 kilometers). You know nothing about this man, the circumference of the Earth, or how one could measure such a thing. As a result, you discard Eratostenes' opinion and (mis)take him for a pretentious loon. But what if other scholars had arrived at very similar measurements, independently of Eratosthenes?

Or even if they had carefully checked his measurement, with a critical eye? Wouldn't that give you enough ground to believe not only that the estimates might be correct, but also that Eratosthenes and his fellow scholars must be quite bright, to be able to achieve such a feat as measuring the Earth?

In this article, we explore how, under some circumstances, we should, and we do infer that a group of individuals whose answers converge are likely to be correct, and to be competent in the relevant area, even if we had no prior belief about either what the correct answer was, or about these individuals' competence.

We begin by reviewing existing studies showing that people infer that competent informants who converge in their opinions are likely to be accurate. The wisdom of crowds literature provides normative grounds for this inference. We then argue that both the experimental and theoretical literature have paid little attention to extending this inference to cases in which there is no information about the informants' competence, and to inferences about the competence of the informants. We first develop normative models, both analytically and with simulations, to show that inferences from convergence to accuracy and to competence are warranted under a wide range of parameters. Second, we present a series of experiments in which participants evaluate both the accuracy and competence of informants as a function of how much their answers converge on a given problem, in the absence of any priors about these individuals' competence, or what the correct answer is.

5.2 Do people infer that individuals whose answers converge tend to be right, and to be competent?

The literature on the wisdom of crowds has treated separately situations with continuous answers, such as the weight of an ox in Galton's famous observation (Galton 1907), and with categorical answers, as when voters have to choose between two options, in the standard Condorcet Jury Theorem (De Condorcet 2014). The continuous and the categorical case are typically modeled with different tools, and they have usually been studied in different empirical literatures (see below). Given that they both represent common ways for answers to converge more or less (e.g. when people give numerical estimates vs. vote on one of a limited number of options), we treat them both here, with different simulations and experiments.

In the continuous case, the most relevant evidence comes from the literature on 'advice-taking' (for review, see, Kämmer et al. 2023). In these experiments, participants are called 'judges' who need to make numerical estimates—sometimes on factual knowledge, e.g. 'What year was the Suez Canal opened first?' (Yaniv 2004), sometimes on knowledge controlled by the experimenters, e.g. 'How many animals were on the screen you saw briefly?' (Molleman et al. 2020). To help answer these questions, participants are given estimates from others, the 'advisors'.

Most of this literature is irrelevant to the point at hand since participants are presented with single estimates, either from a single advisor (e.g. Bednarik and Schultze 2015; Soll and

Larrick 2009; Yaniv 2004; Yaniv and Kleinberger 2000; Harvey and Fischer 1997), or as an average coming from a group of advisors (e.g. Jayles et al. 2017; Mannes 2009), but without any information about the distribution of initial estimates, so that we cannot tell whether participants put more weight on more convergent answers.

Some advice-taking studies provide participants with a set of individual estimates. One subset of these studies manipulates the degree of convergence between groups of advisors, through the variance of estimates (Molleman et al. 2020; Yaniv, Choshen-Hillel, and Milyavsky 2009), or their range (Budescu and Rantilla 2000; Budescu et al. 2003; Budescu and Yu 2007). These studies find that participants are more confident about, or rely more on, estimates from groups of advisors that converge more.

Other studies manipulated the degree of convergence within a group of advisors. These studies present participants with a set of estimates, some of which are close to each other, while others are outliers (Harries, Yaniv, and Harvey 2004; Yaniv 1997, study 3 & 4). These studies find that participants discount outliers when aggregating estimates.

Studies on advice taking thus suggest participants believe that more convergent opinions are more likely to be correct. None of these studies investigated whether participants also believe that those whose opinions converge are also more likely to possess greater underlying competence.

In categorical choice contexts, there is ample and long-standing (e.g. Crutchfield 1955) evidence from experimental psychology that participants are more likely to be influenced by majority opinions, and that this influence is stronger when the majority is larger, both in absolute and in relative terms (e.g., Morgan et al. 2012; for review, see Mercier and Morin 2019). This is true even if normative conformity (when people follow the majority because of social pressure rather than a belief that the majority is correct) is unlikely to play an important role (e.g. because the answers are private, see Asch 1956). Similar results have been obtained with young children (e.g. Fusaro and Harris 2008; Corriveau, Fusaro, and Harris 2009; Bernard et al. 2015; Bernard, Proust, and Clément 2015; Chen, Corriveau, and Harris 2013; Herrmann et al. 2013; Morgan, Laland, and Harris 2015).

If many studies have demonstrated that participants tend to infer that more convergent answers are more likely to be correct, few have examined whether participants thought that this convergence was indicative of the informants' competence. One potential exception is a study with preschoolers in which the children were more likely to believe the opinion of an informant who had previously been the member of a majority over that of an informant who had dissented from the majority (Corriveau, Fusaro, and Harris 2009). However, it is not clear whether the children thought the members of the majority were particularly competent, since their task—naming an object—was one in which children should already expect a high degree of competence from (adult) informants. This result might thus indicate simply that children infer that someone who disagrees with several others on how to call something is likely wrong, and thus likely less competent at least in that domain.

5.3 What inferences from convergence should we expect people to draw?

Should we expect that people be able to infer that more convergent answers likely indicate not only more accurate answers, but also that those who gave the answers were competent? In order to make the best of communicated information, humans have to be able to evaluate it, so as to discard inaccurate or harmful information, while accepting accurate and beneficial information (J. M. Smith and Harper 2003). It has been argued that a suite of cognitive mechanisms—mechanisms of epistemic vigilance—evolved to serve this function (Sperber et al. 2010; Mercier 2020). Since the opinion of more than one individual is often available to us, there should be mechanisms of epistemic vigilance dedicated to processing such situations. It would be these mechanisms that lead us to put more weight on an opinion that is shared by a larger majority (in relative or absolute terms), and, in some cases at least, to discount majority opinion when the opinions haven’t been formed independently of each other (Mercier and Miton 2019). Evidence suggests that these mechanisms rely on heuristics which become more refined with age (Morgan, Laland, and Harris 2015), and which are far from perfect (in particular, they ignore many cases of informational dependencies, see, e.g. Yousif, Aboody, and Keil 2019; Mercier and Miton 2019). As mentioned above, in the experiments evaluating how people process convergent information, the participants had grounds to believe that the information came from competent informants. However, the same mechanisms could lead people to perform the same inference when they do not have such information (especially since, as is shown presently, such an inference is warranted).

Regarding the inference from convergence to competence, other cognitive mechanisms allow us to infer how competent people are, based on a variety of cues, from visual access (Pillow 1989), to the time it takes to answer a question (Richardson and Keil 2022). One of the most basic of these mechanisms infers from the fact that someone was right, that they possess some underlying competence (e.g. Koenig, Clément, and Harris 2004). As a result, if participants infer that convergent opinions are more likely to be accurate, they should also infer that the informants who provided the opinions are competent. Before testing whether participants infer accuracy and competence from convergence, we show that these inferences are normatively warranted, both in the categorical and in the continuous case.

5.3.1 Analytical argument

Regarding the analytical answer, the question can be broken down into two questions. First, can we infer that a population of informants whose answers converge more is, on average, more competent? Second, can we infer that, within this population, individuals who are closer to the consensus or to the average answer are more competent?

In the continuous case, let us imagine a population of informants. Individual opinions are drawn from a normal distribution, centered on the correct answer (i.e., informants have no

systematic bias, as for instance in Galton’s classic demonstration, Galton 1907). The variance of this distribution represents the individuals’ competence: the larger the variance, the lower the competence. We observe the individual answers. In this setting, it is well known in statistics (see also Electronic Supplementary Materials, ESM) that the sample mean (i.e. the mean of the answers of all the informants) is the best estimator of the correct answer, and the sample variance (i.e. the mean squared distance between the answers and the sample mean) is the best estimator of the population’s average competence (best understood here as the least volatile estimator). This means that a population of informants whose answers converge more (lower sample variance) is, on average, more competent (informants tend to answer with a lower variance).

We can extend this argument from populations to individuals: Consider that competence varies within a population, such that each informant’s answer is drawn from their own distribution – always centered on the correct answer, but with different variances. In this case, the distance between each informant’s answer and the sample mean provides the best estimate of that informant’s competence (i.e. individuals whose answers are further away from the sample mean tend to be less competent).

In the categorical case, we define the competence of an informant as the probability of choosing the correct answer. The law of large numbers implies that the relative size of the majority – the share of informants who choose the most chosen answer – is, when the population is large enough, a good approximation of the average competence of the population (e.g. if the average competence is .66, and there are enough informants, then approximately 66% of informants will select the right answer). For smaller populations, the relationship holds, with some degree of noise. In other words, the more the answers converge, the more the population can be inferred to be competent (and the larger the population, the more reliable the inference).

Now, can we infer that informants belonging to the majority are more competent? To do so, we must assume some distribution of competence in the population. Using Bayes theorem, we can then show (see ESM) that the more informants agree with a focal informant, the more the focal informant can be inferred to be competent (Figure 5.1). We also find that, the more competence varies in the population, the more the degree of convergence is indicative of an informant’s competence. For example, if competence is roughly uniform in the population, then being part of a minority likely reflects bad luck, rather than incompetence. More generally, the fact that the focal individual is right (their ‘accuracy’) can be inferred more strongly than their competence, as there is noise in the answer choice (an incompetent individual can always pick the correct answer by chance and vice versa). Figure 5.1) provides an example with a specific categorical choice scenario, under two different population distributions of competence.

5.3.2 Simulations

In order to better understand the influence of different parameters (e.g. degree of convergence, number of individuals) on the relationship between convergence, accuracy, and competence of

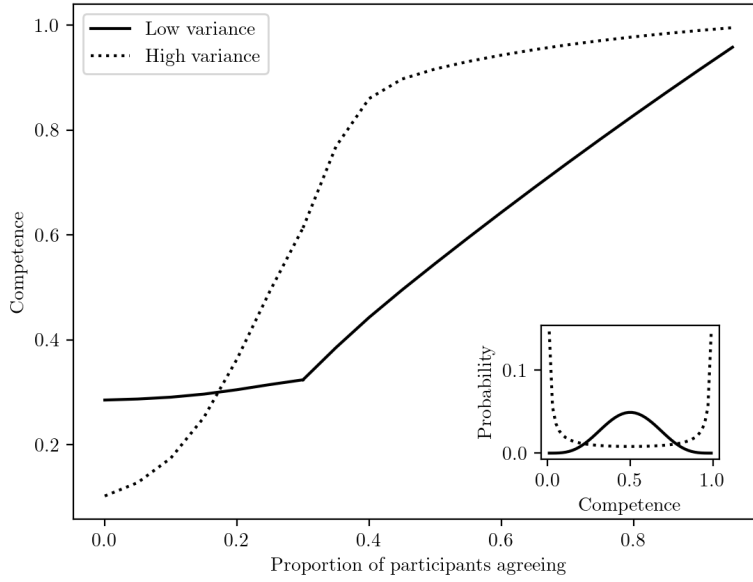


Figure 5.1: Results of the analytic argument. The figure shows the average estimated competence for a focal individual, depending on the proportion of individuals agreeing with them, in a categorical choice scenario. Here, we assume that 20 individuals independently pick one out of 5 choice options. Two situations are represented: one in which there is low variance in the distribution of competence in the population (the bell-shaped curve; solid line), and one in which there is high variance in this distribution (the u-shaped curve; dotted line).

informants, we conducted simulations. In all simulations, we assume agents to be unbiased and independent in their answers, but varying in their competence. All code and data regarding the simulations can be found on Open Science Framework [project page](https://osf.io/6abqy/?view_only=42632bccea604fd7928dbe58e087d23b) (https://osf.io/6abqy/?view_only=42632bccea604fd7928dbe58e087d23b).

5.3.2.1 Continuous case

Groups of agents, with each agents' competence varying, provide numerical answers. We measure how accurate these answers are, and how much they converge (i.e. how low their variance is). We then look at the relationship between convergence and both the accuracy of the answers and the competence of the agents.

More specifically, agents provide an estimate on a scale from 1000 to 2000 (chosen to match the experiments below). Each agent is characterized by a normal distribution of possible answers. All of the agents' distributions are centered around the correct answer, but their standard deviation varies, representing varying degrees of competence. The agents' standard deviation varies from 1 (highest competence) to 1000 (lowest competence). Each agent's competence is drawn from a population competence distribution, expressed by a beta distribution, which can take different shapes. We conducted simulations with a variety of beta distributions which cover a wide range of possible competence populations (see Figure 5.2 A).

A population of around 990000 agents (varying slightly as a function of group sizes) with different competence levels is generated. An answer is drawn for each agent, based on their respective competence distribution. The accuracy of this answer is defined as the squared distance to the true answer. Having a competence and an accuracy value for each agent, we randomly assign agents to groups of, e.g., three. For each group, we calculate the average of the agents' competence and accuracy. We measure the convergence of a group's answers by calculating the standard deviation of the agents' answers. We repeat this process for different sample sizes for the groups, and different competence distributions. Figure 5.2 C displays the resulting relation of convergence with accuracy (left), and competence (right) for different underlying competence distributions and group sizes. We draw broad conclusions from these results after reporting the outcome of the simulations with categorical answers.

5.3.2.2 Categorical case

In the case of categorical answers, convergence can be measured as the relative size of informants picking the same option. The Condorcet jury theorem (De Condorcet 2014; for a recent treatment, see Dietrich and Spiekermann 2013) and its extensions to situations with more than two options (e.g., Hastie and Kameda 2005) already shows that the answer defended by the majority/plurality is more likely to be correct. Regarding competence, Romeijn and Atkinson (2011) have shown that when individuals are more competent, the share of the majority vote

tends to increase. However, they have only studied a case with a uniform distribution of competence. By contrast, here, we investigate a wide range of distributions of competence, and we do not assume that all individuals are equally competent, meaning that, from an observed set of answers, we can attribute an average competence to individuals whose answers form a majority/plurality vs. individuals whose answers form a minority.

We simulate agents whose competence varies and who have to decide between a number of options, one of which is correct. Competence is defined as a value p which corresponds to the probability of selecting the right answer (the agents then have a probability $(1-p)/(m-1)$, with m being the number of options, of selecting any other option). Competence values range from chance level ($p = 1/m$) to always selecting the correct option ($p = 1$). Individual competence levels are drawn from the same population competence beta distributions as in the numerical case (see Figure 5.2 A). Based on their competence level, we draw an answer for each agent. We measure an agent’s accuracy as a binary outcome, namely whether they selected the correct option or not. In each simulation around 99900 agents (varying slightly as a function of the group size) are generated, and then randomly assigned to groups (of varying size in different simulations). Within these groups, we first calculate the share of individuals voting for each answer, allowing us to measure convergence. For example, in a scenario with three choice options and three individuals, two might vote for option A, and one for option C, resulting in two levels of convergence, $2/3$ for A and $1/3$ for C. For each level of convergence occurring within a group, we then compute (i) the accuracy (either 1 if the correct option or 0 else), (ii) the average competence of agents. Across all groups, we then compute the averages of these values, for each level of convergence.

We repeat this procedure varying population competence distributions and the size of informant groups, holding the number of choice options constant at $n = 3$ (for simulations with varying choice options, see ESM). Figure 5.2 B shows the average accuracy (left), and the average competence (right) value as a function of convergence, across different underlying competence distributions and group sizes.

The simulations for the numerical and categorical case demonstrate a similar pattern, which can be summarized as follows:

1. Irrespective of group size (and number of choice options) and of the competence distribution, there is a very strong relation between convergence and accuracy: more convergent answers tend to be more accurate.
2. For any group size and any competence distribution, there is a relation between convergence and the competence of the agents: more convergent answers tend to stem from more competent agents. The strength of this relation is not much affected by the number of agents whose answers are converging, but, although it is always positive, it ranges from very weak to very strong depending on the population’s competence distribution.
3. The relation between convergence and accuracy is always much stronger than the relation between convergence and competence of the agents.

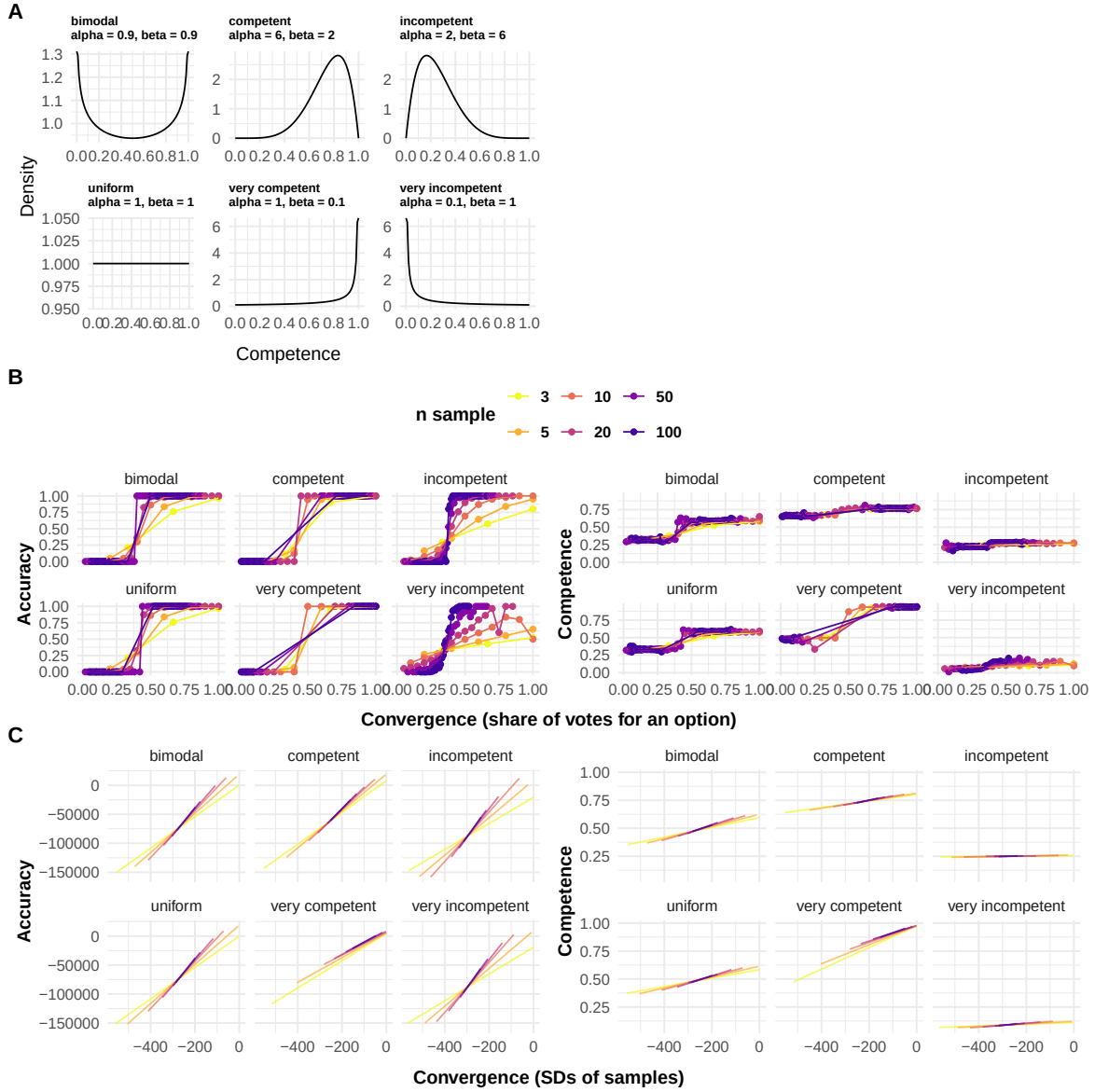


Figure 5.2: Results of the simulations. **A** Shows the different population competence distributions we considered in our simulations. In the continuous simulations, competence values of 0 correspond to a very large standard deviation (1000, with a mean of 1500, on a scale from 1000 to 2000), thereby practically taking the form of a uniform distribution, while competence of 1 corresponds to a very small standard deviation (1, on the same scale). In the categorical simulations, a competence value of 0 corresponds to chance (e.g. in a 3-choice-options scenario, an individual picking the correct answer with a probability of 1/3), while a competence value of 1 corresponds to definitely picking the correct answer. **B** Shows the results of simulations in a categorical setting with three choice options. Points represent average accuracy (left)/competence (right) values by degree of convergence (measured by the share of votes for an option), for different population competence distributions (panels) and sample sizes (colors). **C** Shows the results in a continuous setting. Regression lines represent the correlation between accuracy (left; measured by squared distance to true mean and reversed such that greater accuracy corresponds to being further up on the y-axis) or competence (right), respectively, and convergence (reversed such that greater convergence corresponds to being more right on the x-axis).

5.4 Overview of the experiments

Our models indicate that groups of informants are more likely to have given accurate answers, and to be competent, when their answers converge. In a series of experiments, we test whether people draw these inferences both in numerical tasks (Experiments 1, 2, 3), and in categorical tasks (Experiments 4, 5, 6). By contrast with previous studies, participants were not given any information about the tasks—how difficult they were—and the informants—how competent they might be. There has recently been much interest in investigating whether participants are able to take informational dependencies into account when evaluating convergent information (Yousif, Aboody, and Keil 2019; Mercier and Miton 2019; Desai, Xie, and Hayes 2022; Xie and Hayes 2022; Yin et al. 2024). To test whether participants took into account dependencies between the informants in the present context, this factor was manipulated in two different ways (in Experiment 2, some informants had discussed with each other; in Experiments 3 and 5, some informants had an incentive to provide the same answer). We predicted that dependencies between the informants’ answers should reduce participants’ reliance on the convergence of the answer as a cue to accuracy and to competence.

All experiments were preregistered. All documents, data and code can be found on Open Science Framework [project page](https://osf.io/6abqy/?view_only=42632bccea604fd7928dbe58e087d23b) (https://osf.io/6abqy/?view_only=42632bccea604fd7928dbe58e087d23b). All analyses were conducted in R (version 4.2.2) using R Studio. For most statistical models, we relied on the `lme4` package and its `lmer()` function. Unless mentioned otherwise, we report unstandardized model coefficients that can be interpreted in units of the scales we use for our dependent variables.

5.5 Experiment 1

In Experiment 1, participants were provided with a set of numerical estimates which were more or less convergent, and asked whether they thought the estimates were accurate, and whether the informants making the estimates were competent. Perceptions of accuracy were measured as the confidence in what the participants thought the correct answer was, on the basis of the numerical estimates provided: the more participants think they can confidently infer the correct answer, the more they must think the estimates accurate, on average (the results replicate with a more direct measure of accuracy, see Experiment 3). Our hypotheses were:

H1: When making a guess based on the estimates of (independent) informants, participants will be more confident about their guess when these estimates converge compared to when they diverge.

H2: Participants perceive (independent) informants whose estimates converge more as more competent than informants whose estimates diverge.

We had three research questions regarding the number of informants which report in the ESM.

5.5.1 Methods

5.5.1.1 Participants

We recruited 200 participants from the UK via Prolific (100 female, 100 male; age_{mean} : 39.73, age_{sd} : 15.39, age_{median} : 35.5). Not a single participant failed our attention check. The sample size was determined on the basis of a power analysis for a t-test to detect the difference between two dependent means (“matched pairs”) run on G*Power3. The analysis suggested that a combined sample of 199 would provide us with 80% power to detect a true effect size of Cohen’s $d \geq 0.2$ ($\alpha = .05$, two-tailed).

5.5.1.2 Procedure

After providing their consent to participate in the study and passing an attention check, participants read the following introduction: “Some people are playing games in which they have to estimate various quantities. Each game is different. You have no idea how competent the people are: they might be completely at chance, or be very good at the task. It’s also possible that some are really good while others are really bad. Some tasks might be hard while others are easy. Across various games, we will give you the answers of several players, and ask you questions about how good they are. As it so happens, for all the games, the estimates have to be between 1000 and 2000, but all the games are completely different otherwise, and might require different skills, be of different difficulties, etc. Each player in the game makes their own estimate, completely independent of the others”. After being presented with the results of a game (Figure 5.3), participants had to (i) make a guess about the correct answer based on the estimates they see, “What would you guess is the correct answer, if there is one?”, (ii) estimate their confidence in this guess, “How confident are you that your answer is at least approximately correct?” on a 7-point Likert scale (“not confident at all” to “extremely confident”), (iii) estimate the competence of the group of players whose estimates they saw, “On average, how good do you think these players are at the game?”, also on a 7-point Likert scale (from “not good at all” to “extremely good”).

5.5.1.3 Design

We manipulated two experimental factors, with two levels each: the convergence of the estimates (how close they were to each other; levels: divergent/convergent), and the number of estimates (how many players there were; levels: three/ten). This latter factor was chiefly included to make our results more robust, and was not attached to specific hypotheses. We used a 2 (convergence: divergent/convergent) x 2 (number: three/ten) within-participant design,

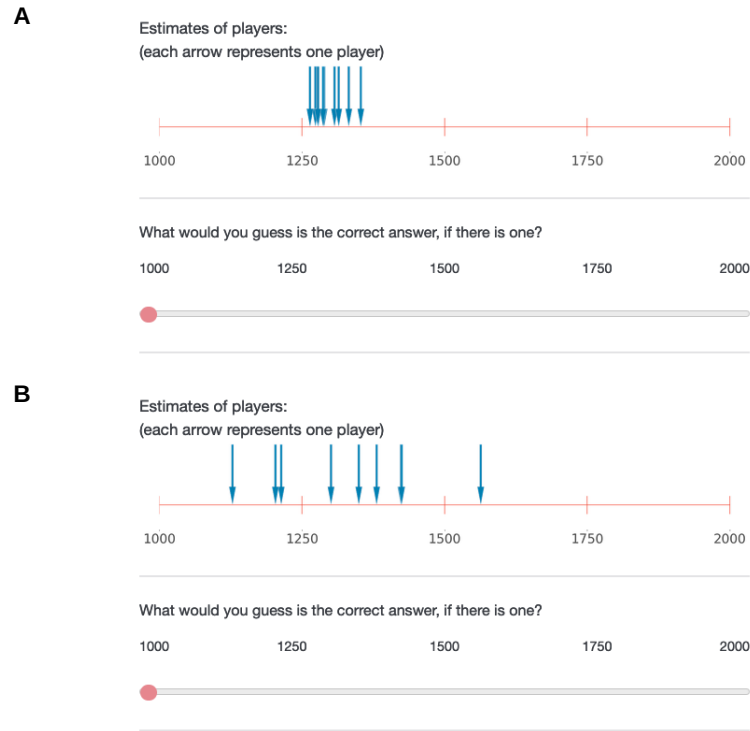


Figure 5.3: Example of two stimuli from Experiment 1, both in the 10 players condition, **A** corresponding to the convergent, **B** to the divergent condition. Similar stimuli are used in Experiments 2 and 3.

with each participant seeing all the conditions. Participants saw two different sets of estimates per condition, for a total of eight sets of estimates per participant.

5.5.1.4 Materials

We generated sets of estimates with random draws from normal distributions. First, we varied the standard deviation of these distributions to simulate the degree of convergence (150 for divergence, 20 for convergence; estimate scale ranged from 1000 to 2000). Second, we varied the number of draws (either three or ten) from these distributions. For each of the four possible resulting conditions, we generated two random draws. We repeated this process for three different sets of estimates, and participants were randomly assigned to one of these sets. More information on how the stimuli were created can be found in the ESM.

5.5.2 Results and discussion

To account for dependencies of observations due to our within-participant design, we ran mixed models, with a random intercept and a random slopes of convergence for participants. In the models for our hypotheses, we control for the number of estimates provided to the participants (three or ten). Visualizations and descriptive statistics can be found in ESM. We find a positive effect of convergence on accuracy: Participants were more confident about their estimate in convergent scenarios (mean = 4.56, sd = 1.448) than in divergent ones (mean = 3.192, sd = 1.392; $\hat{b}_{\text{Accuracy}} = 1.368$ [1.225, 1.51], $p < .001$). We also find a positive effect of convergence on competence: participants rated players as more competent in convergent scenarios (mean = 4.75, sd = 1.237) than in divergent ones (mean = 3.518, sd = 1.266; $\hat{b}_{\text{Competence}} = 1.232$ [1.065, 1.4], $p < .001$).

In an exploratory, non-preregistered analysis, we tested whether the effect of convergence is larger on accuracy than on competence. To this end, we regressed the outcome score on convergence and its interaction with a binary variable indicating which outcome was asked for (accuracy or competence), while controlling for the number of informants. We do not find a statistically significant interaction that would indicate a difference of the effect of convergence ($\hat{b} = 0.135$ [-0.001, 0.271], $p = 0.052$). Pooled across divergent and convergent conditions, we find that participants reported lower perceived accuracy than competence ($\hat{b} = -0.257$ [-0.359, -0.156], $p < .001$).

In summary, as predicted, when the informants' answers were more convergent, participants were more confident that their answers were correct, and they believed the informants to be more competent. This was true both when there were three informants and when there were ten informants.

5.6 Experiment 2

We have shown that it is rational to infer that convergent estimates are more likely to be accurate, and to have been made by competent individuals, only if these individuals were independent and unbiased. However, convergence could come about differently. If the individuals do not make their estimates independently of each other, a single individual might exert a strong influence on the others, making their convergence a poor cue to their accuracy. Alternatively, all individuals might have an incentive to provide a similar, but not accurate answer. In Experiment 2, we investigate the first possibility, and the second in Experiment 3. In particular, for Experiment 2 we rely on past results showing that participants, under some circumstances, put less weight on opinions that have been formed through discussion, by contrast with more independent opinions (Harkins and Petty 1987; see also Lopes, Vala, and Garcia-Marques 2007; Hess and Hagen 2006; Einav 2018). We sought to replicate this finding in the context of convergent estimates, formulating the following hypotheses:

H1: When making a guess based on convergent estimates of informants, participants will be more confident about their guess when informants were independent compared to when they weren't (i.e. they could discuss before).

H2: Participants perceive informants whose estimates converge as more competent when they are independent, compared to when they weren't (i.e. they could discuss before).

Note that these predictions only stem from past empirical results, and are not necessarily normatively justified (modeling the effects of discussion would be beyond the scope of this paper, but see Dietrich and Spiekermann 2024).

5.6.1 Methods

5.6.1.1 Participants

We recruited 200 participants from the UK via Prolific (100 female, 99 male, 1 not-identified; age_{mean} : 40.545, age_{sd} : 13.561, age_{median} : 38.5). Not a single participant failed our attention check. As for experiment 1, the sample size was determined on the basis of a power analysis for a t-test to detect the difference between two dependent means (“matched pairs”) run on G*Power3. The analysis suggested that a combined sample of 199 would provide us with 80% power to detect a true effect size of Cohen’s $d \geq 0.2$ ($\alpha = .05$, two-tailed).

5.6.1.2 Design

In a within-participants design, participants saw both an independence condition, in which they were told “Players are asked to make completely independent decisions – they cannot see

each other’s estimates, or talk with each other before giving their estimates,” and a dependence condition, in which they were told “Players are asked to talk with each other about the game at length before giving their estimates.”

5.6.1.3 Materials

We used the materials generated for the convergent condition of Experiment 1. By contrast to Experiment 1, participants saw only two stimuli in total (one set of estimates per condition), and we only used stimuli involving groups of three informants. Otherwise, we proceeded just as in Experiment 1: we randomly assigned individual participants to one of the three series of stimuli, and for each participant, we randomized the order of appearance of conditions.

5.6.2 Results and discussion

To account for dependencies of observations due to our within-participant design, we ran mixed models, with a random intercept for participants. Visualizations and descriptive statistics can be found in ESM. The data does not support our hypotheses. Participants were slightly less confident about their estimates when the converging informants were independent (mean = 3.775, sd = 1.502), compared to when they discussed (mean = 4.03, sd = 1.389; $\hat{b}_{\text{Accuracy}} = -0.255$ [-0.462, -0.048], $p = 0.016$). The effect is small, but in the opposite direction of what we had predicted. We do not find an effect regarding competence ($\hat{b}_{\text{Competence}} = -0.12$ [-0.272, 0.032], $p = 0.120$).

Contrary to the hypotheses, participants did not deem convergent estimates made after a discussion, compared to independently made estimates, to be less accurate, or produced by less competent individuals. This might stem from the fact that participants, in various situations, neglect informational dependencies (Yousif, Aboody, and Keil 2019), or from the fact that discussing groups actually perform better than non-discussing groups in a range of tasks (for review, see, e.g., Mercier 2016), including numerical estimates (e.g. Mercier and Claidière 2022). As a result, the participants in the current experiment might have been behaving rationally when they did not discount the estimates made after discussion.

5.7 Experiment 3

Experiment 3 tests whether participants are sensitive to another potential source of dependency between convergent estimates: when the individuals making the estimate share an incentive to bias their estimates and disregard accuracy. Even though Experiment 3 is formally similar to Experiment 1, the setting is different, as participants were told that they would be looking at (fictional) predictions of experts for stock values, instead of the answers of individuals in abstract games. In the conflict of interest condition, the experts had an incentive to value the

stock in a given way, while they had no such conflict of interest in the independence condition. We tested for an interaction, namely whether the positive effect of convergence is reduced when informants are systematically biased, compared to when they are not. On this basis, we formulate four hypotheses, two of which are identical to those of Experiment 1, and only apply in the independent condition, and two that bear on the interaction:

***H1a:** Participants perceive predictions of independent informants as more accurate when they converge compared to when they diverge.*

***H1b:** Participants perceive independent informants as more competent when their predictions converge compared to when they diverge.*

***H2a:** The effect of convergence on accuracy (H1a) is more positive in a context where informants are independent compared to when they are in a conflict of interest.*

***H2b:** The effect of convergence on competence (H1b) is more positive in a context where informants are independent compared to when they are in a conflict of interest.*

We have not conducted simulations to validate these predictions. However, given the operationalization chosen, in which convergence should provide little evidence of either accuracy or competence, we believe the predictions regarding the superiority of the independent informants stem naturally from the model of independent informants presented above.

5.7.1 Methods

5.7.1.1 Participants

The interaction design of our third experiment made the power analysis more complex and less standard than for experiments one and two. Because we could build upon data from the first experiment, we ran a power analysis by simulation. The simulation code is available on the OSF, and the procedure is described in the preregistration document. The simulation suggested that 100 participants provide a significant interaction term between 95% and 97% of the time, given an alpha threshold for significance of 0.05. Due to uncertainty about our effect size assumptions and because we had resources for a larger sample, we recruited 199 participants for this study – again, from the UK and via Prolific (99 female, 100 male; age_{mean} : 40.296, age_{sd} : 12.725, age_{median} : 38).

5.7.1.2 Procedure

After providing their consent to participate in the study and passing an attention check, participants read the following introduction: “You will see four scenarios in which several experts predict the future value of a stock. You have no idea how competent the experts are.

It’s also possible that some are really good while others are really bad. As it so happens, in all scenarios, the predictions for the value of the stock have to lie between 1000 and 2000. Other than that, the scenarios are completely unrelated: it is different experts predicting the values of different stocks every time.” Participants then saw the four scenarios, each introduced by a text according to the condition the participant was assigned to. To remove any potential ambiguity about participants’ inferences on the accuracy of the estimates, we replaced the question about confidence to one bearing directly on accuracy: “On average, how accurate do you think these three predictions are?” on a 7-point Likert scale (“not accurate at all” to “extremely accurate”). The question about competence read: “On average, how good do you think these three experts are at predicting the value of stocks?”, also assessed on a 7-point Likert scale (from “not good at all” to “extremely good”).

5.7.1.3 Design

We manipulated two factors: informational dependency (two levels, independence and conflict of interest; between participants) and convergence (two levels, convergence and divergence; within participants). In the independence condition, the participants read “Experts are independent of each other, and have no conflict of interest in predicting the stock value - they do not personally profit in any way from any future valuation of the stock.” In the conflict of interest condition, the participants read “All three experts have invested in the specific stock whose value they are predicting, and they benefit if other people believe that the stock will be valued at [mean of respective distribution] in the future.”

5.7.1.4 Materials

The distributions presented were similar to those of Experiment 1, although generated in a slightly different manner (see ESM). Each participant rated four scenarios, two for each level of convergence. By contrast to Experiment 1, all scenarios only involved groups of three informants.

5.7.2 Results and discussion

To account for dependencies of observations due to our within-participant design, we ran mixed models, with a random intercept and a random slope for convergence for participants. We find evidence for all four hypotheses. As for the first set of hypotheses, to match the setting of experiment one, we reduced the sample of Experiment 3 to half of the participants, namely those who were assigned to the independence condition. On this reduced sample, we ran the exact same analyses as in Experiment 1 and replicated the results. As for accuracy, participants rated informants in convergent scenarios (mean = 5.28, sd = 1.052) as more accurate than in divergent ones (mean = 3.4, sd = 1.08; $\hat{b}_{\text{Accuracy}} = 1.88$ [1.658, 2.102], $p < .001$). As for competence, participants rated informants in convergent scenarios (mean = 5.235, sd = 0.992)

as more competent than in divergent ones (mean = 3.61, sd = 1.111; $\hat{b}_{\text{Competence}} = 1.625$ [1.411, 1.839], $p < .001$).

The second set of hypotheses targeted the interaction of informational dependency and convergence (Figure 5.4). In the independence condition, the effect of convergence on accuracy was more positive ($\hat{b}_{\text{interaction, Accuracy}} = 0.991$ [0.634, 1.348], $p < .001$) than in the conflict of interest condition ($\hat{b}_{\text{baseline}} = 0.889$ [0.636, 1.142], $p < .001$). Likewise the effect of convergence on competence is more positive ($\hat{b}_{\text{interaction, Competence}} = 0.802$ [0.474, 1.13], $p < .001$) than in the conflict of interest condition ($\hat{b}_{\text{baseline}} = 0.823$ [0.591, 1.056], $p < .001$).

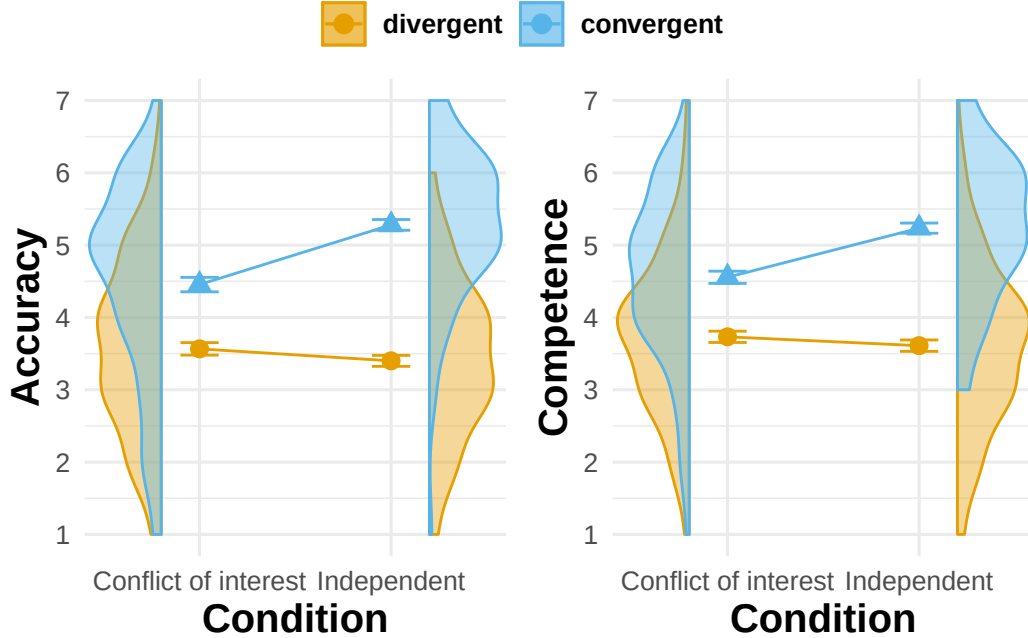


Figure 5.4: Results of Experiment 3, showing the distributions of accuracy and competence ratings by convergence and informational dependency.

In an exploratory, non-preregistered analysis, we tested whether the effect of convergence is larger on accuracy than on competence. To this end, we regressed the outcome score on convergence and its interaction with a binary variable indicating which outcome was asked for (accuracy or competence), while controlling for informational dependency. We find a negative interaction effect, indicating that pooled across independent and conflict of interest conditions, the effect of convergence had as smaller effect on competence than on accuracy ($\hat{b} = -0.161$ [-0.294, -0.028], $= 0.018$). Pooled across all conditions, participants reported higher perceived competence than accuracy ($\hat{b} = 0.108$ [0.025, 0.191], $= 0.011$).

Experiment 3 shows that, when the individuals making the estimates are systematically biased, participants put less weight on the convergence of their estimates to infer that the estimates

are accurate, and that the individuals making them are competent.

5.8 Experiment 4

In a second series of experiments, we test similar predictions to those of the previous experiments, but in a categorical choice context. The set-up is similar to that of Experiment 1, except that the outcomes seen by the participants are not numerical estimates, but choices made between a few options. An additional difference is that participants rate a focal informant, and not a group of informants. There were two reasons for this choice: First, accuracy is not on a continuum as in the first three experiments (an option was either correct or not), so forming an average across informants who chose different options was less sensible. Second, rating a focal individual allowed us to have a minority condition, which would not have been possible when providing an average rating for a group. Experiment 4 tests hypotheses that are analogous to those of Experiment 1:

H1: Participants perceive an estimate of an independent informant as more accurate the more it converges with the estimates of other informants.

H2: Participants perceive an independent informant as more competent the more their estimate converges with the estimates of other informants.

5.8.1 Methods

5.8.1.1 Participants

We ran a power simulation to inform our choice of sample size. All assumptions and details on the procedure can be found on the OSF. We ran two different power analyses, one for each outcome variable. We set the power threshold for our experiment to 90%. The power simulation for accuracy suggested that even for as few as 10 participants (the minimum sample size we simulated data for), we would have a power of close to 100%. The simulation for competence suggested that we achieve statistical power of at least 90% with a sample size of 30. Due to uncertainty about our assumptions and because it was within our budget, we recruited 100 participants, from the UK and via Prolific (50 female, 50; age_{mean} : 37.32, age_{sd} : 11.526, age_{median} : 36).

5.8.1.2 Procedure

After providing their consent to participate in the study and passing an attention check, participants read the following introduction: “To be able to understand the task, please read the following instructions carefully: Some people are playing games in which they have to select the correct answer among three answers. You will see the results of several of these

games. Each game is different, with different solutions and involving different players. All players answer independently of each other. At first, you have no idea how competent each individual player is: they might be completely at chance, or be very good at the task. It's also possible that some players are really good while others are really bad. Some games might be difficult while others are easy. Your task will be to evaluate the performance of one of the players based on what everyone's answers are." They were then presented to the results of eight such games (Figure 5.5). To assess perceived accuracy, we asked: "What do you think is the probability of player 1 being correct?". Participants answered with a slider on a scale from 0 to 100. To assess perceived competence, we asked participants: "How competent do you think player 1 is in games like these?" Participants answered on a 7-point Likert scale (from (1) "not competent at all" to (2) "extremely competent").

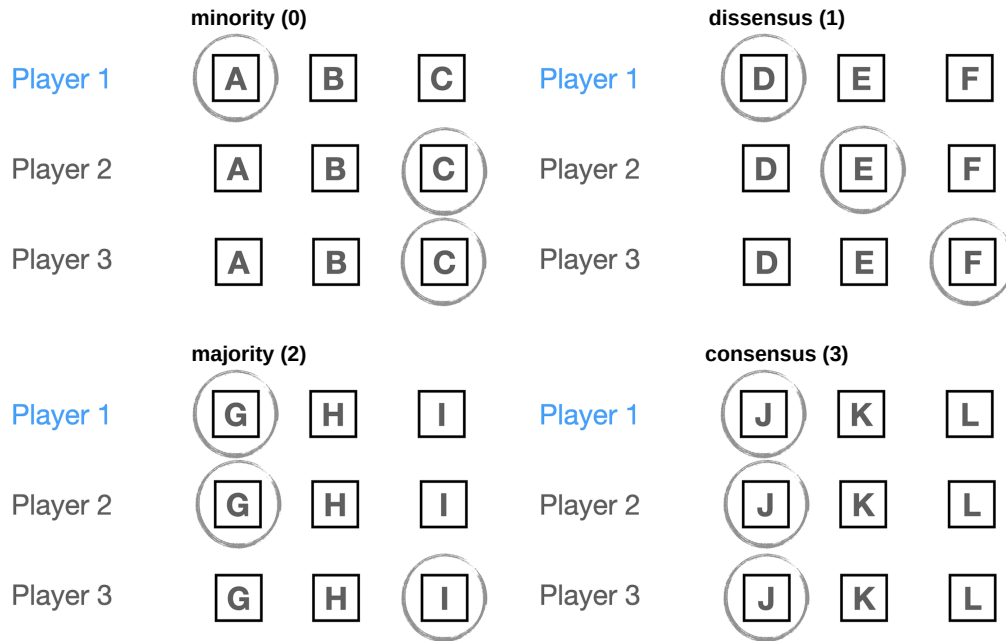


Figure 5.5: One set of stimuli by level of convergence, in Experiment 4 (similar stimuli are used in Experiments 5 and 6). A full set of stimuli can be found in the ESM.

5.8.1.3 Design

We manipulated convergence within participants, by varying the ratio of players choosing the same response as a focal player (i.e. the one that participants evaluate). The levels of convergence are: (i) consensus, where all three players pick the same option [coded value = 3]; (ii) majority, where either the third or second player picks the same option as the first player [coded value = 2]; (iii) dissensus, where all three players pick different options [coded value

= 1]; (iv) minority, where the second and third player pick the same option, but one that is different from the first player’s choice [coded value = 0]. In our analysis, we treat convergence as a continuous variable, assigning the coded values in squared parenthesis here.

5.8.1.4 Materials

All participants saw all four conditions, with two stimuli per condition. Each participant therefore saw eight stimuli in total (4 convergence levels x 2 stimuli).

5.8.2 Results and discussion

To account for dependencies of observations due to our within-participant design, we ran mixed models, with a random intercept and a random slope for participants.

As in the numerical setting, we found a positive effect of convergence on both accuracy ($\hat{b}_{\text{Accuracy}} = 16.838$ [15.009, 18.668], $p < .001$; on a scale from 0 to 100) and competence ($\hat{b}_{\text{Competence}} = 0.683$ [0.578, 0.788], $p < .001$; on a scale from 1 to 7).

In the ESM, we show that compared to what the normative models would predict, participants underestimate the effect of convergence on both accuracy and competence, but especially on accuracy.

In an exploratory, non-preregistered analysis, we tested whether the effect of convergence is larger on accuracy than on competence. To do so, we first standardized both outcome scores to account for the different scales. We then regressed the outcome score on convergence and its interaction with a binary variable indicating which outcome was asked for (accuracy or competence). We find a negative interaction, indicating that convergence had a smaller effect on competence than on accuracy ($\hat{b} = -0.095$ [-0.136, -0.053], $< .001$; units in standard deviations).

5.9 Experiment 5

Experiment 5 is a conceptual replication of Experiment 3 in a categorical instead of a numerical case: are participants less likely to infer that more convergent estimates are more accurate, and the individuals who made them more competent, when the estimates are made by individuals with a conflict of interest pushing them to all provide a given answer, compared to when they are made by independent individuals? The independence condition of Experiment 5 also serves as a replication of Experiment 4, leading to the following hypotheses:

H1a: Participants perceive an estimate of an independent informant as more accurate the more it converges with the estimates of other informants.

H1b: Participants perceive an independent informant as more competent the more their estimate converges with the estimates of other informants.

H2a: The effect of convergence on accuracy (H1a) is more positive in a context where informants are independent compared to when they are biased (i.e. share a conflict of interest to pick a given answer).

H2b: The effect of convergence on competence (H1b) is more positive in a context where informants are independent compared to when they are biased (i.e. share a conflict of interest to pick a given answer).

5.9.1 Methods

5.9.1.1 Participants

We ran a power simulation to inform our choice of sample size. All assumptions and details on the procedure can be found on the OSF. We ran two different power analyses, one for each outcome variable. We set the power threshold for both to 90%.

The power simulation for accuracy suggested that for 80 participants, we would have a power of at least 90% for the interaction effect. The simulation for competence suggested that with already 40 participants, we would detect an interaction, but only with 60 participants would we also detect an effect of convergence. Due to uncertainty about our assumptions and because resources were available for a larger sample, we recruited 200 participants, in the UK and via Prolific (99 female, 100, 1 non-identified; age_{mean} : 41.88, age_{sd} : 13.937, age_{median} : 39).

5.9.1.2 Procedure

After providing their consent to participate in the study and passing an attention check, participants read the following introduction: “We will show you three financial advisors who are giving recommendations on investment decisions. They can choose between three investment options. Their task is to recommend one. You will see several such situations. They are completely unrelated: it is different advisors evaluating different investments every time. At first you have no idea how competent the advisors are: they might be completely at chance, or be very good at the task. It’s also possible that some are really good while others are really bad. Some tasks might be difficult while others are easy. Your task will be to evaluate the performance of one of the advisors based on what everyone’s answers are.” To assess perceptions of accuracy, we asked: “What do you think is the probability of advisor 1 making the best investment recommendation?”. Participants answered with a slider on a scale from 0 to 100. To assess perceptions of competence, we asked: “How competent do you think advisor 1 is regarding such investment recommendations?” Participants answered on a 7-point Likert scale (from (1)“not competent at all” to (7)“extremely competent”).

5.9.1.3 Design

We manipulated convergence within participants, and conflict of interest between participants. In the conflict of interest condition, experts were introduced this way: “The three advisors have already invested in one of the three options, the same option for all three. As a result, they have an incentive to push that option in their recommendations.” Participants assigned to the independence condition read: “The three advisors are independent of each other, and have no conflict of interest in making investment recommendations.”

5.9.1.4 Materials

We used the same stimuli as in Experiment 4. Identical to Experiment 4, participants saw all four convergence conditions, with two stimuli (i.e. expert predictions) per condition. Each participant therefore saw eight stimuli in total (4 convergence levels x 2 stimuli).

5.9.2 Results and discussion

To account for dependencies of observations due to our within-participant design, we ran mixed models, with a random intercept and a random slope for convergence for participants.

We find evidence for all four hypotheses (see Figure 5.6). To test H1a and H1b, we use the same analyses as in Experiment 4, restricted on the independence condition, and replicate the results. We find a positive effect of convergence on both accuracy ($\hat{b}_{\text{Accuracy}} = 12.337$ [10.362, 14.311], $p < .001$) and competence ($\hat{b}_{\text{Competence}} = 0.562$ [0.459, 0.665], $p < .001$).

The second set of hypotheses targeted the interaction of informational dependency and convergence (Figure 5.6). In the independence condition, the effect of convergence on accuracy was more positive ($\hat{b}_{\text{interaction, Accuracy}} = 3.008$ [0.027, 5.988], $p = 0.048$) than in the conflict of interest condition ($\hat{b}_{\text{baseline}} = 9.329$ [7.232, 11.426], $p < .001$). Likewise, the effect of convergence on competence was more positive ($\hat{b}_{\text{interaction, Competence}} = 0.165$ [0.014, 0.316], $p = 0.032$) than in the conflict of interest condition ($\hat{b}_{\text{baseline}} = 0.397$ [0.291, 0.503], $p < .001$).

In an exploratory, non-preregistered analysis, we tested whether the effect of convergence is larger on accuracy than on competence. To do so, we first standardized both outcome scores to account for the different scales. We then regressed the outcome score on convergence and its interaction with a binary variable indicating which outcome was asked for (accuracy or competence), while controlling for informational dependency. We find a negative interaction, indicating that convergence had a smaller effect on competence than on accuracy ($\hat{b} = -0.079$ [-0.113, -0.044], $p < .001$; units in standard deviations).

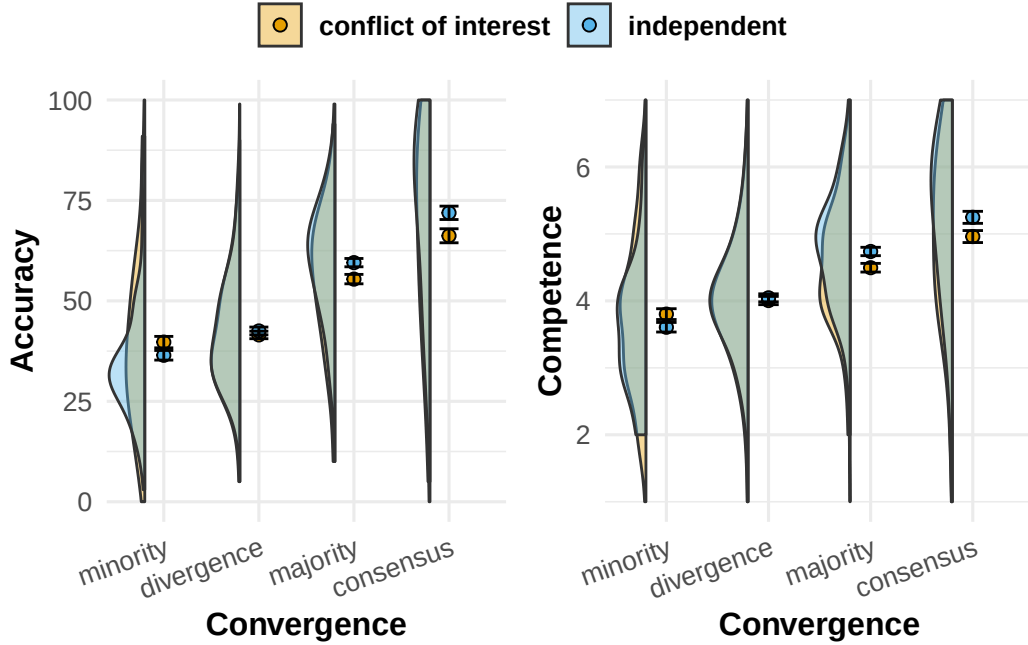


Figure 5.6: Interaction of convergence and informational dependency.

5.10 Experiment 6

Experiment 6 is a replication and extension of Experiment 4 in which we test the effect of the number of choice options (three and ten, instead of only three). Our simulations suggested that, at least for some underlying population competence distributions, consensus should be more indicative of competence when there are more choice options, compared to fewer (see ESM).

First, considering only the three options condition, we ran a direct replication of experiment 4. Second, following the results from our model, we predict that

H1: The effect of convergence on accuracy (H1a) is more positive in a context when informants can choose among ten response options compared to when they can choose among only three.

H2: The effect of convergence on competence (H1b) is more positive in a context when informants can choose among ten response options compared to when they can choose among only three.

5.10.1 Methods

5.10.1.1 Participants

We ran a power simulation to inform our choice of sample size. All assumptions and details on the procedure can be found on the OSF. We used previous experiments and estimates of our models to inform our choice of parameter values. We ran two different power analyses, one for each outcome variable. We set the power threshold for our experiment to 90%. The power simulation for accuracy suggested that for 140 participants we would cross the power threshold of 90% for the interaction effect (power = 0.928). The simulation for competence suggested that with 300 participants, we would detect an interaction with a power of 87%. Due to budget constraints, we considered aiming for a sample of 300 participants as good enough, although slightly below our threshold. Due to two failed attention checks, our final sample consisted of 298 subjects, recruited, as in all experiments, in the UK and via Prolific (149 female, 149; age_{mean} : 42.091, age_{sd} : 13.065, age_{median} : 40).

5.10.1.2 Procedure

We used the same procedure as in Experiment 4, with the addition of one condition described below.

5.10.1.3 Design

The number of choice options was manipulated between participants. Participants were randomly assigned to either see stimuli with three options (as in Experiment 4), or stimuli with ten options. Participants assigned to the ten options condition were divided into one of two distinct sub-conditions: one in which the range of the answers corresponds to the range of the three options condition, and another with increased range (see ESM). We found no differences between the two sub-conditions and collapsed them into a single ten options condition.

5.10.1.4 Materials

For the three options condition, we used the same stimuli as in Experiments 4 and 5. For the ten options condition, we created new sets of stimuli (see ESM). Identical to Experiments 4 and 5, participants saw all four convergence conditions, with two stimuli per condition. Each participant therefore saw eight stimuli in total.

5.10.2 Results and discussion

To account for dependencies of observations due to our within-participant design, we ran mixed models, with a random intercept and a random slope for convergence for participants.

We replicate the results of experiment 4, but do not find evidence for an interaction between convergence and the number of choice options. To match the setting of experiment one, we reduced the sample to half of the participants, namely those who were assigned to the three options condition. On this reduced sample, we ran the exact same analyses as in experiment 4 and replicated the results. We find a positive effect of convergence on both accuracy ($\hat{b}_{\text{Accuracy}} = 15.679$ [14.112, 17.246], $p < .001$) and competence ($\hat{b}_{\text{Competence}} = 0.65$ [0.564, 0.736], $p < .001$). This finding also holds across the entire sample, pooling three and ten choice option observations ($\hat{b}_{\text{Accuracy}} = 16.305$ [15.124, 17.485], $< .001$; $\hat{b}_{\text{Competence}} = 0.675$ [0.611, 0.739], $< .001$).

We do not find evidence of an interaction, i.e. evidence that the number of choice options changes the effect of convergence ($\hat{b}_{\text{interaction, Accuracy}} = 1.252$ [-1.11, 3.613], $= 0.298$; $\hat{b}_{\text{interaction, Competence}} = 0.05$ [-0.078, 0.178], $= 0.442$).

We tested whether the effect of convergence is larger on accuracy than on competence in an exploratory, non-preregistered analysis. To this end, we first standardized both outcome scores to account for the different scales. We then regressed the outcome score on convergence and its interaction with a binary variable indicating which outcome was asked for (accuracy or competence), while controlling for the number of choice options. In line with all previous experiments, we find a negative interaction, indicating that convergence had a smaller effect on competence than on accuracy ($\hat{b} = -0.049$ [-0.075, -0.023], $< .001$).

Experiment 6 replicates Experiments 3 and 4 (independence condition), but suggests that the number of options the informants choose from does not powerfully affect participants' estimates of the informants' accuracy or competence. In the ESM, we show that, compared to our model, participants underestimate the effect of the number of choice options.

5.11 General discussion

Using both analytical arguments and simulations, we have shown that, under a wide range of parameters, more convergent answers tend to be more accurate, and to have been made by more competent informants.

In two experiments (Experiment 1, and independence condition of Experiment 3), we find that participants presented with a set of more (rather than less) convergent numerical estimates find the estimates more accurate, and the individuals making the estimates more competent, thus drawing normatively justified inferences. Experiment 2 suggests that participants do not think that a discussion between the individuals makes their convergence less indicative of

accuracy or their competence. By contrast, Experiment 3 reveals that, when the individuals making the estimates are systematically biased by a conflict of interest, participants put less weight on the convergence of their estimates.

Similar results are obtained in a categorical choice context, in which participants see the answers of individuals made within a limited set of options. Experiments 4, 5 (independence condition), and 6 show that, the more the answers converge, the more participants believe them to be accurate and the individuals who made to be competent, again drawing normatively justified inferences. Experiment 5 shows that these inferences are weakened when the convergence can be explained by a conflict of interest (as in Experiment 3). Experiment 6 fails to find an effect of the number of options.

We also observe that, in line with our simulations, participants draw stronger inferences from convergence to accuracy than to competence in five of the six experiments.

On the whole, participants thus appear to draw normatively justified inferences: 1. They infer that more convergent answers are more accurate; 2. They infer that more convergent answers are coming from more competent informants; 3. The inference on accuracy tends to be stronger than the inference on competence ; 4. Both the inference on accuracy and on the inference on competence are weaker when the informants have a conflict of interest, but not when they are merely discussing with each other. The only exception appears to be that participants do not take into account the number of options in categorical choices scenarios.

5.12 Conclusion

When people see that others agree with each other, they tend to believe that they are right. This inference has been evidenced in several experiments, both for numerical estimates (e.g. Molleman et al. 2020; Yaniv, Choshen-Hillel, and Milyavsky 2009; Budescu and Rantilla 2000; Budescu et al. 2003; Budescu and Yu 2007), and for categorical choices (e.g., Morgan et al. 2012; for review, see Mercier and Morin 2019). However, these experiments do not test whether this inference of accuracy of the information extends to an inference of competence of the informants. Moreover, by their design, participants arguably assumed a degree of competence among the informants. For instance, when children are confronted with several individuals who agree on how to name a novel object (e.g. Corriveau, Fusaro, and Harris 2009), they can assume that these (adult) individuals tend to know what the names of objects are. If a certain competence of the informants is assumed, then well-known results from the literature on judgment aggregation—the wisdom of crowds—show that the average opinion of a set of individuals is, in a wide range of circumstances, more likely to be accurate than that of a single individual (e.g. Larrick and Soll 2006).

Here, we assumed no prior knowledge of individual competence, asking the question: if we see informants, whose competence is unknown, converge on an answer, is it rational to infer that this answer is more likely to be correct, and that the informants are likely to be competent?

We have shown that the answer is yes on both counts—assuming there is no systematic bias among the informants. An analytical argument and a series of simulations revealed that, for both the numerical choice context and the categorical choice context, the more individuals agree on an answer, the more likely the answer is correct, and the more likely the individuals are competent, with the former effect being stronger than the latter. Moreover, this is true for a wide range of assumed population distributions of competence. In a series of experiments, we have shown that participants (UK) draw these inferences, but that they do so less when there is reason to assume a bias among informants.

The results—both simulations and experiments—are a novel contribution to the wisdom of crowds literature. In this literature—in particular that relying on the Condorcet Jury Theorem—a degree of competence is assumed in the individuals providing some answers. From that competence, it can be inferred that the individuals will tend to agree, and that their answers will tend to be accurate. Here, we have shown that the reverse inference—from agreement to competence—is also warranted, and that it is warranted under a wide range of circumstances: If one does not suspect any systematic bias, convergence alone can be a valid cue when determining who tends to be an expert. This finding qualifies work suggesting that people need a certain degree of expertise themselves, in order to figure out who is an expert (Nguyen 2020; Hahn, Merdes, and Sydow 2018).

The present experiments show that people draw inferences from convergence to accuracy and to competence, but then do not precisely show how they do it. As suggested in the introduction, it is plausible that participants use the combination of two heuristics: one leading them from convergence to accuracy, and one from accuracy to competence. This two step process is coherent with the observation that the inference from convergence to accuracy (one step) is stronger than that from convergence to competence (two steps). However, our results are also compatible with participants performing two inferences from convergence, one to accuracy and one to competence, the latter being weaker because the inferences roughly follow the normative model. Our results are coherent with work on other mechanisms of epistemic vigilance that process the combination of several informants’ opinions, showing that these mechanisms rely on sensible cues, but also systematically fail to take some subtle cues into account (Mercier and Morin 2019).

A most prominent context in which the inferences uncovered here might play a role is that of science. Much of science is counterintuitive, and most people do not have the background knowledge to evaluate most scientific evidence. However, science is, arguably, the institution in which individuals end up converging the most in their opinions (on consensus being the defining trait of science by contrast with other intellectual enterprises, see Collins 2002). For instance, scientists within many disciplines agree on things ranging from the distance between the solar system and the center of the galaxy to the atomic structure of DNA. This represents an incredible degree of convergence. When people hear that scientists have measured the distance between the solar system and the center of the galaxy, if they assume that there is a broad agreement within the relevant experts, this should lead them to infer that this measure is accurate, and that the scientists who made it are competent. Experiments have already

shown that increasing the degree of perceived consensus among scientists tends to increase acceptance of the consensual belief (Van Stekelenburg et al. 2022), but it hasn't been shown yet that the degree of consensus also affects the perceived competence of scientists.

In the case of science, the relationship between convergence and accuracy is broadly justified. However, at some points of history, there has been broad agreement on misbeliefs, such as when Christian theologians had calculated that the Earth was approximately six thousand years old. To the extent that people were aware of this broad agreement, and believed the theologians to have reached it independently of each other, this might have not only fostered acceptance of this estimate of the age of the Earth, but also a perception of the theologians as competent.

The current study has a number of limitations. In our simulations, we assume agents to be independent and unbiased. Following previous work generalizing the Condorcet Jury Theorem to cases of informational dependency (Ladha 1992), more robust simulations would show that—while still assuming no systematic bias—our results hold even when agents influence each others' answers. Regarding our experiments, if the very abstract materials allow us to remove most of the priors that participants might have, they might also reduce the ecological validity of the results. Although the main results replicate well across our experiments, and we can thus be reasonably certain of their robustness, it's not clear how much they can be generalized. Experimental results with convenience samples can usually be generalized at least to the broader population the samples were drawn from—here, UK citizens (Coppock 2019). However, we do not know whether they would generalize to other cultures.

These limitations could be overcome by replicating the present results in different cultures, using more ecologically valid stimuli. For instance, it would be interesting to test whether the inference described here, from convergence to competence, might be partly responsible for the fact that people tend to believe scientists to be competent (Cologna et al. 2024). Finally, future studies could also attempt to systematically model cases of informational dependencies more subtle than those used in Experiments 3 and 5, to derive and test normative predictions.

5.12.0.1 Data availability

Data for all experiments and the simulations is available on the OSF project page (https://osf.io/6abqy/?view_only=42632bccea604fd7928dbe58e087d23b).

5.12.0.2 Code availability

The code used to create all results (including tables and figures) of this manuscript is also available on the OSF project page (https://osf.io/6abqy/?view_only=42632bccea604fd7928dbe58e087d23b).

5.12.0.3 Competing interest

The authors declare having no competing interests.

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